syngo.via Operator Manual

Operator Manual – Basics

VB40A

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SIEMENS Healthineers

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Operator Manual – Basics

VB40A

Legend

	Indicates a hint Is used to provide information on how to avoid operating errors or information emphasizing important details
	Indicates the solution of a problem Is used to provide troubleshooting information or answers to frequently asked questions
●	Indicates a list item
✓	Indicates a prerequisite Is used for a condition that has to be fulfilled before starting a particular operation
◆	Indicates a one-step operation
1 2 3	Indicates steps within operating sequences
Italic	Is used for references and for table or figure titles
→	Is used to identify a link to related information as well as previous or next steps
Bold	Is used to identify window titles, menu items, function names, buttons, and keys, for example, the Save button
Orange	Is used to emphasize **particularly** important sections of the text
Courier	Is used for on-screen output of the system including code-related elements or commands
`Courier`	Is used to identify inputs you need to provide
Menu > Menu Item	Is used for the navigation to a certain submenu entry
<variable>	Is used to identify variables or parameters, for example, within a string
⚠ **CAUTION**	CAUTION Used with the safety alert symbol, indicates a hazardous situation which, if not avoided, could result in minor or moderate injury or material damage. CAUTION consists of the following elements: • Information about the nature of a hazardous situation • Consequences of not avoiding a hazardous situation • Methods of avoiding a hazardous situation

WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.

WARNING consists of the following elements:

- Information about the nature of a hazardous situation
- Consequences of not avoiding a hazardous situation
- Methods of avoiding a hazardous situation

WARNING

Legend

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1 Introduction

syngo.via is a post-processing and viewing software that features worklist management, automatic preprocessing, as well as postprocessing tools.

This document gives an overview on the main functionality in *syngo*.via. It also introduces you to the way of working with the *syngo*.via software. Detailed information is available in the Online Help.

For better readability, we refer to the user in the masculine form.

1.1 Documentation overview

The *syngo*.via software offers several levels of user assistance, beginning with tooltips and extended tooltips on the screen linking into the comprehensive *syngo*.via Online Help. The *syngo*.via software is also accompanied by the *syngo*.via Basic Operator Manual and the *syngo*.via Administrator Manual.

Both manuals constitute the Instructions for Use of *syngo*.via. They are available in local languages as online-version.

Some functions described in the documentation may not be available on your system. Contact your Clinical Administrator or Siemens Healthineers for more information.

1.1.1 Administrator Manual

The *syngo*.via Administrator Manual contains the safety advisories (regarding administration), and provides information about administration and configuration of your *syngo*.via system. It is available in local languages.

Detailed information is available in the *syngo*.via Administration Online Help.

Most configuration tasks of *syngo*.via are performed with the *syngo*.via Administration Portal. For detailed information about the User Interface and the corresponding configuration tasks, see (→ *"Configuration in syngo.via" in the Online Help*).

1.1.2 Basic Operator Manual

The *syngo*.via Basic Operator Manual contains the safety advisories, gives you an overview of the most important tools that are available in your *syngo*.via system, and contains introductory as well as basic information. The *syngo*.via Basic Operator Manual is available in local languages.

1.1.3 Online Help

The comprehensive information for *syngo*.via is the context-based Online Help. You can quickly access the Online Help following the links in an extended tooltip, by pressing the **F1** key (on the client), or by clicking the Question Mark icon on the access bar (both on the client and in the *syngo*.via Administration Portal).

The information range depends on your licenses.

The Online Help is also available as a website. You can access it with a browser with the URL "http://<*syngo*.via-server>:8090", where <*syngo*.via-server> is the IP address or the host name of your *syngo*.via server.

The Online Help is available for the standard user interface languages.

1.1.4 Supported languages

syngo.via supports the following user interface languages:

- English
- German
- French
- Spanish

- Japanese
- Chinese (simplified)

The *syngo*.via Administration Portal is available in English only.

syngo.via does not support local differences in languages, for example, Spanish (Mexico) is displayed the same as Spanish (Spain).

If the Online Help is not available in your language, the Online Help as well as the tooltips are displayed in English.

1.1.5 Further documentation

- **Data Sheet**

  Detailed technical data is provided in the *syngo*.via Data Sheet, VB40.

- **Applications and disease-specific workflows**

  Applications and disease-specific workflows are accompanied by their own documentation and safety advisories. The Online Help system comprises the help modules for workflows and applications installed on your individual *syngo*.via system. See (➔ *Application Online Help*).

  Contact your Clinical Administrator or Siemens Healthineers for more information.

- **e-Clips**

  These application videos are available in the Online Help as a supplement for the user documentation.

- **MM Reading Quick Guide**

  This Quick Guide Online Help takes you on a tour through the *syngo*.via workflow MM Reading. This Online Help contains descriptions of the main functions of this workflow as well as the basics for getting started quickly.

  The MM Reading Quick Guide is available in English only.

- **Pre-Installation Manual for server virtualization**

  This manual describes the installation procedure of a VMware Container to use *syngo*.via as a virtual machine.

- **Reporting Adapter**

  This manual describes the *syngo*.via Reporting Adapter to integrate *syngo*.via reports into external reporting systems.

1.1.6 Comments and suggestions

We hope that this documentation gives you the support you need. The documentation is regularly updated along with the software development, and your comments will be taken into account to improve the documentation.

Please address your comments or suggestions to your local Siemens Healthineers representative.

We look forward to receiving your feedback.

2 Intended use

syngo.via is a software solution intended to be used for viewing, manipulation, communication, and storage of medical images.

It can be used as a stand-alone device or together with a variety of cleared and unmodified *syngo* based software options.

syngo.via supports interpretation and evaluation of examinations within healthcare institutions, for example, in Radiology, Nuclear Medicine, and Cardiology environments.

The system is not intended for the displaying of digital mammography images for diagnosis in the U.S.

2 Intended use

3 Legal notes

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3.1 User profiles

Please note that the following profiles may vary in practice depending on (hospital) organization, qualification, and personal responsibilities and can only be considered as a general guide.

The following user roles and profiles have been identified for *syngo*.via:

Technologist (Radiographer)

- Role
  - Quality assurance of the images
  - Preparation of images for reading, such as manipulation of size, position, or windowing values
  - Creation of preliminary markers or measurements
  - Generation of additional result series, for example, parallel ranges
  - Sending data for archiving
- Knowledge and experience
  - Vocational High School
  - Radiation technologist training
  - Work experience
  - Product training certificate or comparable product experience

Radiologist

- Role 1 (for example, residents)
  - Performing the imaging examination
  - Exporting of patient images
  - Completing but not verifying reports
  - Intervention, responsible for the safety of patients and employees
  - Sending data for archiving
- Role 2 (for example, attending physicians)
  - Tasks of **Role 1** listed above
  - Justifying indication (medical check)
  - Verifying reports

- Knowledge and experience
  - Doctor of Medicine (MD)
  - Specialist for radiology
  - Work experience
  - Product training certificate or comparable product experience

Clinical Administrator

- Role
  - Configuration of application-related settings
  - Configuration of workflow-related settings
  - Customization of client software options
  - Patient data administration
  - First support contact for radiologists
- Knowledge and experience
  - Radiologist, or Radiographer, with work experience, product training certificate or comparable product experience, and knowledge of the RIS/PACS workflow
  - Application specialist who typically works in the radiology department
  - Archiving, management of Short-Term Storage (STS)
  - DICOM knowledge

  In addition to the information provided in the *Configuration Online Help*, the clinical administrator needs to follow the information in the *Administration Online Help*.

IT Administrator

- Role
  - Management of the IT infrastructure of the clinical network and the PACS/RIS system
  - Administration and configuration of IT components of the *syngo*.via system
  - Responsibility for data security, data protection, and license handling
  - Configuration and administration of DICOM nodes
  - Management of Short-Term Storage (STS), as well as backup and archiving
  - User account and role management
- Knowledge and experience
  - Expert IT knowledge of hardware, software, and operating-system issues
  - English language skills (fluent in spoken and written English)
  - Data management
  - User administration
  - Networks
  - RIS/PACS workflow
  - License handling

3.2 Education and training

To empower your staff with expertise, Siemens Healthineers offers continuous education tailored to your institutions needs based on a blended learning approach.

At the time your *syngo*.via is installed, an initial training is provided to guarantee a seamless start with your *syngo*.via software.

The handover training is delivered by the Clinical Education Specialist and includes the following:

- Clinical integration of the main modality
- Pre-training clarification (remote or on-site)
- Dedicated number of *syngo*.via training hours/days, depending on the specific applications and users for your institution

To give you the possibility to increase your knowledge at your pace, you have access to the Siemens Healthineers learning platform **PEPconnect**, where you can benefit from online training and educational videos, focused on the utilization of your *syngo*.via clinical applications. https://pep.siemens-info.com

With **PEPconnections** – our premium subscription for workforce education management – you can assign and manage the education of your staff, ensuring to your employees a continuous education path.

In addition, an **extended education plan** can be provided which may consist of remote trainer or classroom training.

For further information about education plans, please contact your local Siemens Healthineers sales representative.

For general information on classroom training, please see https://training.healthcare.siemens.com.

3.3 Open Source Software

The product contains, among other things, Open Source Software developed by third parties.

The Open Source Software used in the product and the license agreements concerning this software can be found in the ReadMeOSS. These Open Source Software files are protected by copyright. Your compliance with those license conditions will entitle you to use the Open Source Software as foreseen in the relevant license.

In the event of conflicts between Siemens Healthineers license conditions and the Open Source Software license conditions, the Open Source Software conditions shall prevail with respect to the Open Source Software portions of the software. The Open Source Software is licensed royalty-free.

Insofar as the applicable Open Source Software license conditions provided for this, you can order the source code of the Open Source Software from your Siemens Healthineers sales contact – against payment of the shipping and handling charges – for a period of at least 3 years since purchase of the product.

We are liable for the product including the Open Source Software contained in it pursuant to the license conditions applicable to the product. Any liability for the Open Source Software beyond the program flow intended for the product is explicitly excluded.

Furthermore, any liability for defects resulting from modifications to the Open Source Software by you or third parties is excluded.

We do not provide any technical support for the product if it has been modified.

4 Safety advisory

Warnings indicate a potential hazard to the health or life of patients or personnel.

Cautions indicate a hazardous situation which, if not avoided, could result in minor or moderate injury.

The following cautions are valid for the *syngo*.via basic installation. Consider also the safety advisories of the applications and the disease-specific workflows that are available with your *syngo*.via installation.

In addition to the following cautions, administrators need to consider the safety advisory from the *syngo*.via Administrator Manual.

General advice for the U.S.A.

For the U.S.A., please note the following:

- Monitors (displays) should not be used for diagnosis unless the used monitor (printer) has specifically received 510(k) clearance for this purpose.
- Paper printouts should not be used for diagnosis unless the used Post Script printer has specifically received 510(k) clearance for this purpose

4.1 Online Help viewer

CAUTION

Contents (such as PDFs or videos) uploaded by customers/ users to the Online Help viewer can be mixed up with Siemens Healthineers provided content.

Possible injury of the patient or wrong diagnosis.

- Content uploaded by users is explicitly marked with an upload symbol in the Online Help viewer. Use the instructions for use attached to / provided by / accompanying the Siemens Healthineers medical device as your primary source of information.

4.2 Reporting support

CAUTION

syngo.via implements ACR Reporting and Data Systems (RADS) for reporting imaging findings.

The diagnosis and treatment may be based on incorrect information.

- Verify all system generated results for RADS findings results using your clinical knowledge. System generated results can be overwritten when needed.

CAUTION

Unexpected report findings and text due to copy-paste errors, imported or automatically filled-in data.

The diagnosis and treatment may be based on incorrect information.

- Read the report text carefully before signing it off or storing it.

CAUTION

User or referring physician treats printed report as official report although it has not been signed off. The reporting application does not provide any digital sign-off functionality such as a RIS.

The diagnosis and treatment may be based on outdated or preliminary information.

- Printed reports are only valid with a signature. Always check reports for validity.

CAUTION

Dates and numbers in snapshot images can be displayed in different formats than dates in the *syngo*.via user interface and in reports.

Unclear dates and values, for example, mix-up of day and month representation like DD/MM instead of MM/DD.

- Be aware that unexpected formats may be used for dates and values in snapshots. The *syngo*.via client uses the default regional settings of the operating system. Do not customize this default regional setting with a different date format.

CAUTION

Findings may be omitted after selecting another report template.

Delayed diagnosis due to missing findings.

- Drag & drop appropriate sections to the template or switch back to the initial template to integrate already existing findings again.

CAUTION

Stored reports that belong to a certain workflow are not automatically updated after the workflow has been reopened and changed, for example, due to new finding information, changed patient identification attributes, or medical alerts/ allergies.

Outdated information in reports.

- After performing changes in the reopened workflow, a new report versions must be created to reflect these changes by pressing the reporting icon in the patient tab.

CAUTION

Patient-specific information left in clipboard

Patient privacy is compromised.

- Close the work item of the patient when leaving the workstation, especially if the workstation is in a public place accessible to multiple users.

CAUTION

Insecure storage of patient reports.

Patient privacy is compromised

- Only store reports on secure systems with controlled access.

4.3 Emergency concept

CAUTION

Computers are not infallible and unexpected errors may occur. In addition, scheduled downtimes are necessary to maintain the software.

The system may not be available for use, for instance, in the operating room or for an emergency case.

- Work out an emergency plan for response to non-availability of the system or the network, for instance, to use a system on a different network or to use print-outs or films.

CAUTION

There is no mechanism to recognize an emergency patient in *syngo*.via.

The user may not be aware of a pending emergency case.

- Establish and apply a method to identify emergency patients, for instance, by adding a corresponding notice to the patient identification.

4.4 Image evaluation

CAUTION

Use of inappropriate devices (displays, printers) to review and report radiological images.

Incorrect review of images.

- Review and reporting of images require an optimum display of images.
- Only use suitable monitors and approved DICOM printers for review and reporting of images.
- Follow the maintenance and care instructions given in the manufacturer's documentation.

CAUTION

Evaluation and visualization of results may become imprecise because of display size, round-off errors, subpixel precision, zooming, and other factors.

Inaccurate measurement results.

- Be aware of precision limitations when using calculated measurement values.
- Only perform measurements on images displayed in a ratio of 1:1 (1 display pixel = 1 acquired pixel).

CAUTION

The display of slightly different data sets (e.g. prior/current study) makes it difficult to distinguish the data sets from each other when switching between them. Image data displayed may be not as expected when opening saved workflows.

The diagnosis and treatment may be based on incorrect information.

- Customize and read image text to ensure differentiation of the different data sets.

  Use the study date as the primary identification source.

CAUTION

Patient demographic data in image text may be completely hidden by user configuration.

Wrong diagnosis due to a mix-up of patients especially in patient merge scenarios where different patients are shown in one layout.

- Always apply the full image text to check the identification of the patient in patient merge scenarios.

CAUTION

Images processed with radial ranges may be displayed in oblique orientation.

Wrong diagnosis due to orientation mix-up.

- Set up a departmental policy for creation of radial ranges regarding the orientation and review of the resulting outcome.

CAUTION

2D reference line may show an anatomical position that does not correspond exactly to the referenced images. This can be caused by e.g. patient movement or patient breathing as the same table position (one common frame of reference) is used for all displayed image series.

The diagnosis and treatment may be based on incorrect information.

- Be aware that the accuracy of the 2D reference line with respect to the shown anatomy may deviate due to external influences.

CAUTION

Findings are not stored in *syngo*.via if the workflow is opened in limited write access mode.

Loss of intermediate results

- Always open the workflow with full write access if you want to process your findings in the findings navigator or in the reporting application.

CAUTION

Image resolution and quality of preview images provided in the Patient Browser are not sufficient for clinical diagnosis.

The diagnosis and treatment may be based on incorrect information.

- When performing clinical diagnosis only use the high resolution images in the imaging area of *syngo*.via.

CAUTION

A combined report is used for results from different workflows, for example, tumor measurements of the same region in MM Onco and MR Onco.

Wrong tumor burden calculation and wrong tumor trending in case of follow-up examinations.

- If your tumor burden calculation is based on measurements performed in different workflows, ensure that no lesion has been measured twice with a RECIST measurement.
- Check the number of lesions calculated for tumor burden.

4.5 Lossy compressed images or data

CAUTION

Lossy compressed data is not sufficient for primary diagnosis.

The diagnosis and treatment may be based on insufficient information.

- Do not use lossy compression if images are to be used for further diagnosis.

  This can be checked in the export profile and the system configuration.
- Do not use lossy compressed images for primary diagnosis. A compression factor is shown on compressed images.

  Such images are marked by “CQ!”, “CR!” or “CRx”, where x states the compression ratio. The compression indicator is only displayed, if image text is enabled and compression rate is configured to be visible.
- Do not use videos for primary diagnosis, which use a lossy codec.

4.6 Loading of images

CAUTION

Additional images may arrive at the system while reading a patient.

Diagnosis may be made with incomplete data.

- Check the list/thumbnail view in the **Series** panel for additional images before completing the diagnosis.

CAUTION

Patient mix-up is possible if data from different patients are open in different applications.

Diagnosis and treatment may be based on incorrect information.

- Check the patient name in each application.

CAUTION

syngo.via provides a mode that allows users to load local studies for an examination as priors, based on a site-specific unique Enterprise Master Patient Index (EMPI).

Wrong diagnosis because the EMPI for a certain patient is not unique.

- EMPI mode is disabled by default and should only be configured if the unique Enterprise Master Patient Index (EMPI) for each patient in the DICOM attribute OtherPatientID (0010,1000) can be ensured by the site. EMPI mode can only be configured by the Customer Care Center.
- Always consider that prior patient identification attributes displayed in the image text and in the patient tab on the access bar may differ, for example, the patient name or the patient ID may not be the same. Use the Other Patient ID value in the Patient Browser to verify that the current study and the prior study belong to the same patient.
- EMPI is only supported for studies assigned as priors from the short-term storage (STS). Searches based on an Enterprise Master Patient Index are not supported for pre-fetching studies and querying/retrieving studies.
- If the EMPI option is enabled but causes issues with prior loading, contact the Customer Care Center immediately to have the EMPI settings adjusted.

4.7 Importing and exporting data

CAUTION

Incorrect measurement results due to incompletely imported or received studies.

Wrong diagnosis

- Always check that studies have been completely imported to or received by the system.
- If an error occurred during import, try to import the study again.
- If the error occurs again, decide whether the incomplete data poses a risk of wrong diagnosis and must be deleted.

CAUTION

Not all servers in the multi-server environment did respond to a user query.

The diagnosis and treatment may be based on incomplete information.

- When starting a query in a multi-server environment, always consider the warning icon that is displayed on the status bar if not all servers respond.

CAUTION

Removing the CD, DVD, or memory stick before completion of the writing process renders the disk unreadable.

Data loss due to incomplete writing of storage medium.

- Do not open CD or DVD drive until a message appears informing that writing is complete.
- Do not remove a memory stick before performing safe removal of the hardware.

CAUTION

Low-quality storage media are used.

Data loss due to improper handling or low-quality CDs or DVDs.

- Use only quality media approved for medical applications when storing patient data.
- Observe proper handling guidelines for CDs and DVDs.

4.8 Correcting and rearranging of patient data

CAUTION

The active patient could contain data from other patients or series (for example, after user interactions, data import from CD/DVD, or merge from RIS).

The diagnosis and treatment may be based on incorrect information.

- Check that the data refer to the active patient before using the data.

CAUTION

External systems (such as the PACS) respond in an unpredictable way if already sent data is subsequently corrected or rearranged at *syngo*.via.

Inconsistent data, as re-archiving can fail and DICOM objects can be mixed-up

- Always delete the affected DICOM objects from the external system first.
- After that, manually resend the corrected data from *syngo*.via.

CAUTION

There is no direct undo option for correct or rearrange operations available on the UI.

Studies remain assigned to the wrong patient

- In general, perform correct and rearrange operations carefully.
- In case a rearrangement was performed by mistake, the wrongly assigned study can be reassigned to the original patient in the same way. Alternatively, if you have closed the **Correction of Study** dialog already, search for the wrongly assigned study in the result list of the Patient Browser (**Corrected** flag is set) and perform the rearrange operation again.
- In case a correction was performed by mistake, the original data can be retrieved from an external source such as PACS or RIS. Alternatively, the wrongly corrected study can be exported and, afterwards, the original information is contained in DICOM tag (0400,0561). There, it can be accessed with an external DICOM viewer.

CAUTION

The **Correct and Rearrange** functionality for patient data is limited to the single-server mode and is not synchronized with other servers in a multi-server installation.

Patient data and patient identification may be outdated or mixed up; patient data may not be found.

- On a multi-server client, always verify where data for a patient is located.
- Perform the corrections on each server manually, using a dedicated single-server client.
- Delete the data on all servers and resend the data.
- Configure an HL7 message for all servers and use the RIS for correction.

4.9 System configuration

CAUTION

Configurable automatic rules can become complex.

Unexpected system behavior or loss of data due to definition of complex automatic rules.

- Test all new rules to ensure that the results conform to your expectations.

CAUTION

Use of an anti-virus software that is not provided by Siemens Healthineers.

Malicious software can cause harms to the system and loss of all patient data.

- The administrator is responsible for the configuration of the anti-virus software.
- Configure and regularly update the anti-virus software.

CAUTION

The installation of other non-system software interferes with the system software.

Harms to the server system and loss of all patient data.

- The installation of software not provided from Siemens Healthineers is not allowed on the server system.
- Problems arising due to interactions with non-Siemens Healthineers software are not the responsibility of Siemens Healthineers.

4.10 Data storage and archiving

CAUTION

Data transfers between systems are not guaranteed.

Loss of data if data are deleted locally before they have been successfully transferred to another system.

- Only systems and communication protocols supporting Storage Commitment guarantee successful transfer to another system.
- In other transfers, it is necessary to verify the correct arrival of the data at the remote system itself.
- Do not delete local data until the transfer to the remote system is verified.

CAUTION

Long-term archiving is not provided with this system.

Loss of data (including reports) can occur if data have not been transferred to a long-term archive.

- Verify that local data has been transferred successfully to a long-term archive, before deleting it.
- Check the status bar for archiving failures (indicated by an error icon).
- Check the **Job View** to locate the error and take appropriate measures (for example, re-start failed archive jobs).

4.11 Image post-processing

CAUTION

Automatic image registration of two data sets (for example, PET-CT) is not infallible.

The diagnosis and treatment may be based on incorrect information.

- Check the automatic registration of images before making a diagnosis.
- Adjust the image registration manually, if necessary.

CAUTION

The evaluation of distances, angles, and ROIs in 3D (VRT, MIP, DRR or SSD) and projection (plain-film X-ray or fluoroscopic) images can be inaccurate.

Diagnosis may be wrong because of incorrect measurement results.

- Do not use projection images for critical measurements.

CAUTION

The volume evaluated may be larger than the volume drawn by the user due to round-off errors and sub-voxel precision.

Inaccurate measurement results

- Be aware of precision limitations when using VOI measurement tools (freehand/sphere). Please consider this effect when interpreting measurement results.

CAUTION

Manual masking, bone and table removal mask out tissue (e.g. vessels, stents, or plaques), leave bone fragments, or hide other diagnostically relevant information.

The diagnosis is based on incomplete or wrong information.

- Use the review and edit mode to verify the masking using the parameters for the threshold, size and noise.
- Add relevant items to or remove unwanted parts from the bone mask.
- Make sure that the correct body region is selected for bone removal or fracture analysis.
- Check the results for consistency with the original 2D image.

CAUTION

The automatic localization and labeling of vertebrae and/or ribs may be incorrect.

Incorrectly labeled vertebrae / ribs can result in a wrong diagnosis or treatment.

- Manually check, correct (if required) and confirm the accuracy of all labels (spine labels, left and right rib labels) using your clinical knowledge as a radiologist.

  Hint: Note that any automatically localized but unlabeled vertebrae / ribs are marked with an "???" and should be corrected.

CAUTION

Vertebral label positions may be wrong in multiple image series as the positions share a common acquisition frame of reference. Rib label positions may be wrong in multiple image series as the positions are calculated in each series separately.

Wrong diagnosis possible due to a mix-up of vertebrae / ribs.

- Always check the correctness of the label positions in each labeled series within a study.

CAUTION

After label edit operations, the assignment of the vertebrae or the ribs may be wrong.

Wrong diagnosis possible due to mix-up of vertebrae / ribs.

- Always check that after label propagation and delete operations, the spine / rib labels still correspond to your clinical knowledge as a radiologist.

CAUTION

Use of a black and white monitor for the **CT Lung Change** tool where anatomy changes are not displayed in color.

Wrong diagnosis is possible due to missing color information.

- Only use the **CT Lung Change** tool with color monitor.

CAUTION

Anatomical change visualization between selected studies is incorrect.

Wrong diagnosis is possible if results from CT Lung Change tool is used directly for diagnosis.

- The **CT Lung Change** tool is only a visual aid for getting the user attention towards changes. Diagnosis should be finalized only after reading the images and making quantifications on the actual anatomy.

CAUTION

When the filter is turned on, only a part of the identified CAD markers will be displayed.

User misses to read/review all identified CAD markers.

- Be aware that if the filter is enabled, only a subset of the identified CAD markers will be displayed. An icon indicates that the CAD marker list is filtered. Please take this into account before completing your review of the identified CAD markers.

4.12 Printing

CAUTION

Hard copies could contain data from several examinations of a patient or even from different patients. If the attributes that identify the examination or patient are not displayed, patients may be mixed up or the examinations of an individual patient may be mixed up.

The diagnosis and treatment may be based on incorrect information.

- Check that the information on the hard copy contains attributes that identify the examination or the patient before using the printout for diagnosis.

CAUTION

Result of true size in printout can depend on printer settings, round-off errors and other factors.

True Size printout does not correlate exactly to the real anatomy size.

- Be aware of precision limitations when printing in **True Size**. Always compare the image scalebar within the printed images for validation with a physical measurement unit such as a ruler to ensure that the printout has the real anatomy size.
- If your layout contains small segments because of which the scalebar cannot be applied, either validate true size directly by measuring the printed anatomy or choose a different layout.

CAUTION

For paper printers, the “printed” flag is set after the images have been successfully transferred but it cannot be guaranteed that the images have actually been printed out.

No paper printout is available for the transferred images.

- Make sure that the paper printout is actually available before you delete the corresponding image data.

4 Safety advisory

5 Getting started

This section describes how you can quickly start working with your *syngo*.via software:

- (➔ Page 47 *Logging on to syngo.via*)
- (➔ Page 48 *Logging on to a specific server*)
- (➔ Page 49 *Closing syngo.via*)
- (➔ Page 50 *Basic operation in the Viewer*)

5.1 Logging on to *syngo*.via

Depending on the security policy at your site, you can directly log on to *syngo*.via as the logged on Windows user (single sign-on) or you have to enter further credentials.

After logging on to *syngo*.via, your client connects to the default server. Image data is retrieved from this server.

To log on to another server, see: (➔ Page 48 *Logging on to a specific server*)

✓ The computer is started and you are logged on to Windows.

1. Double-click the **syngo.via – Single Sign On** icon to log on to *syngo*.via without further authentication as the logged on Windows user.

   – or –

   Double-click the **syngo.via Client** icon.

   The **Logon** dialog box is opened.

2. Enter your *syngo*.via **User Name** and **Password**. Note that the password is case-sensitive.
3. You might have to select a **Domain**. After initial log on, the domain is remembered by the system.

4 Click **OK** to confirm.

For information how to switch the logged on user, see: (➔ *"Switching the user" in the Application Online Help*)

Administrators can configure if users have to log on separately with their credentials.

Only if the Enterprise license is available, your client can connect in multiserver mode. (➔ *Logging on to syngo.via in multiserver mode*)

5.2 Logging on to a specific server

You can log on to *syngo*.via and select the server that *syngo*.via is connected to. The **Patient Browser** will receive the patient data from this server. You can set the selected server as the default server that syngo.via logs on by default.

You can connect your client PC to a specific server.

Communication between a client and a server requires identical software versions.

When you connect to another server than your Home server, other settings may apply than specified at the Home server.

Some client or user-specific configurations like monitor configuration, UI language, layouts, and community links (also automatic UI scaling) are stored at the server. These configurations might have to be adapted.

You can open several *syngo*.via clients in parallel, each on a different monitor, connected to a specific server. (For example, for a two-monitor configuration, one *syngo*.via client on the right and one on the left monitor.) Therefore, you need to configure monitors when you are connected to a specific server. See (➔ *Configuration Online Help*)

1 Double-click the **syngo.via - Server Selection** icon on your desktop.

The **syngo.via Server Selection** dialog box is opened.

2 In the **Server** field, enter a server name, IP address, or host name.

– or –

Select the desired server, IP address, or host name from the list. The first list entry represents the last connected server. List entries may be generated by an automatic launch from a RIS or a PACS.

You can cut, copy, paste, and delete entries in the **Server** list using the context menu or the Windows standard shortcut keys.

3 To make the selected server the Home server that *syngo*.via connects by default, select the **Set as home Server** check box.

4 Select **Login with the credentials of your operating system**.

– or –

Select **Login with other credentials**.

5 Click the **Connect** button.

The **syngo.via Logon** dialog box is opened.

6 In case of using other credentials, enter your **User Name** and **Password**. Note that the password is case-sensitive.

7 Click the **OK** button.

5.3 Closing *syngo*.via

◆ Click the **Close** icon at the upper right to close the *syngo*.via client software.

Please note the following:

- If you have to leave the workplace unattended, be sure to lock the computer.

  At a locked computer, only the user who has logged in to Windows can unlock the workplace.
- If you are leaving the workplace for a longer period, make sure to log off the *syngo*.via application.
- If you are leaving the workplace for a longer period without logging off, *syngo*.via logs off automatically. Workflows that are opened in the Viewer are closed and *saved*.

Important information for *syngo*.via mobile devices:

syngo.via declines automatic sleep of the operating system as long as it is running.

The timeout for the sleep modus of the operating system starts after *syngo*.via has been closed.

5.4 Basic operation in the Viewer

The information in this section helps you to work with the Viewer. It explains how to access important functions and shows how to use the mouse and the keyboard.

5.4.1 Mouse interactions

In *syngo*.via, you can use a mouse with a scroll wheel or a mouse with three mouse buttons. By default you can perform the following actions using the mouse buttons and the scroll wheel. The description below refers to a mouse with two mouse buttons and a scroll wheel.

Mouse interactions in the image area

The right mouse button allows you to navigate; the left mouse button lets you use the selected tool. The middle mouse button/ scroll wheel is used for windowing or scrolling. Modifier keys such as the **Alt** key or the **Ctrl** key enable you to perform more functions with the mouse:

	Uses the currently selected tool By default, **Zoom/Pan** is activated for 2D images and for 3D images.
Ctrl +	Swap segment Click and drag an image to another segment. The dragged image swaps position with the image in the target segment.
	On double-click, the segment is enlarged (blow up) or reduced again.
	Opens the context menu
	Windows the selected image, when you press the mouse wheel: • Move the mouse up / down to decrease / increase the brightness • Move the mouse to the right / left to decrease / increase the contrast • Double-click to reset the windowing values (MM Reading only)
Ctrl +	Windows the selected images: Move the mouse up / down to decrease / increase the brightness Move the mouse to the right / left to decrease / increase the contrast

	Keep the mouse button pressed to scroll through original (stack or stripe) or calculated 2D images (MPR, MIP Thin, VRT Thin, MPR/MPR).
	Rotates a VRT/MIP volume in the selected segments
	Use the scroll wheel to scroll through images of the selected segments
Alt +	Use the scroll wheel to scroll through time points or data groups of 2D images
	Scrolls through time points or data groups of 2D images
Alt +	Rotates MPR images and 2D images

5.4.2 Keyboard interactions

The keyboard lets you access functions quickly. The keyboard can be used to perform a specific action or to change the behavior of the mouse interactions.

The navigational keys refer to different functions depending if you are in stripe or stack mode and if you display 2D images or 3D images.

Navigation keys in 2D

Arrow keys up/down	Scrolls through the images
Arrow keys up/down	Scrolls through the images in stripe of stacks layouts
Ctrl + Arrow key	Jumps to the next segment
Alt + Arrow key	Scrolls through data groups, if available
Home/End	Jumps to the first/last image in a stack

Navigation keys in 3D

Arrow key	• Rotates a 3D volume • Scrolls through a slice or slab
Ctrl + Arrow key	Jumps to the next segment

Function keys

F1	Opens the Online Help (with information belonging to the situation/workflow)
Shift + F5-F8	Depending on the configuration for the specific modality: • Applies windowing presets • Applies color LUT presets
F9	Toggles between alpha values within a fused segment (MPR/MPR). 0% - currently applied alpha value - 100%
F10	Inverts the color or grayscale values (color LUT) for the images in the selected segments
F11	Maximizes the image area. Pressing again toggles back.

Shortcut keys to access tools and functions

A	Sends all images of a stack from the image area to the film sheet.
Alt (press and hold)	Allows you to move evidence objects even if a tool is currently active.
Ctrl + **A**	Selects all segments
Ctrl + **Alt** + **A**	Closes the workflow and sends the results of the current session to the configured archive
C	Activates the **ROI Circle** tool
D	Activates the **Distance Line** tool
E	Sends the currently selected image from the image area to the film sheet.
F	Creates a clinical finding and lists it in the **Findings Navigator** or **Findings** panel
Ctrl + **F**	Activates the **Quick Search** in the **Patient Browser**
G	Hides or shows graphical objects
H	Hides or shows reference lines
Ctrl + **G**	Selects all graphical objects
I	Activates the **Scroll** tool
Shift + **I**	Activates or deactivates the performance mode in the scrolling without skipping mode Scrolling performance is improved due to image compression
L	Activates the **Pixel Lens** tool
Alt + **L**	Starts and stops the **Layout Editing** mode
M	Activates the **Marker** tool

N	Enables/disables **Lock Navigation**
O	Activates the **Rotate** tool
P	Activates the **Punch** function to draw and punch only one contour (quick punch mode)
Ctrl + **P**	Starts the print job
Q	Removes the **Clip box** and **Clip Plane** in the selected segments
R	Activates the **3D Reference Point** tool
Alt + **R**	Opens the **Report Editor**
S	Creates a snapshot finding out of the selected segments and lists it in the **Findings** panel or **Findings Navigator**
Ctrl + **S**	Adds a snapshot to an existing finding in the **Findings** panel or **Findings Navigator**.
Ctrl + **Alt** + **S**	Closes the workflow without archiving
T	Hides or shows the image text
U	Enables or disables scrolling without skipping when using the right mouse button. If you scroll with skipping enabled, an indicator is displayed in the segment.
V	Activates the **VOI Sphere** tool
W	Activates the **Windowing** tool
Y	Enables or disables **Synchronization** functions
Z	Activates the **Zoom/Pan** tool
Spacebar	Activates the **Movie** tool Starts and stops a movie
Ctrl + **Shift** + **A**	Activates fast correction of alignment

Page Down	Rotates the aligned object clockwise in **Visual Alignment** mode
Standard shortcut keys	
Esc	Deactivates the current tool For volume restriction with planes: • Hides the handles of the planes • Hides the planes
Del	Removes the selected item/overlay
Ctrl + **C**	Copies the selected object to the clipboard
Ctrl + **H**	Resets the settings to the initial state
Ctrl + **X**	Cuts the selected objects
Ctrl + **V**	Pastes an object from the clipboard
Alt + **Tab**	Lets you select another primary window or application
Ctrl +	Selects multiple objects
Shift +	Selects multiple objects in sequence
Ctrl + **Z**	Lets you undo the last actions stepwise. Furthermore, the mini toolbars of most tools have an **Undo** icon.
Ctrl + **Y**	Restores your recently undone actions stepwise. Furthermore, the mini toolbars of most tools have a **Redo** icon.

5.4.3 About corner menus

The corner menus provide tools for image preparation, optimization, evaluation, and property settings.

They are placed in the corners of each segment. They are indicated by a watermark and grouped by functionality, as shown in the following table. The tools offered in the corner menus depend on various conditions, such as current step, and image and display type.

The corner menu can be opened by clicking or hovering the mouse over the segment corner. How to open the corner menu can be configured.

 Top left corner menu	Image manipulation tools (➔ Page 114 *Orientation tools*) (➔ Page 115 *Tools to mask out structures*) (➔ Page 114 *Registration Tools (Fusion)*) (➔ Page 117 *Tools for restricting the volume*) (➔ Page 102 *Scrolling tools*) (➔ Page 105 *Optimization and preparation tools*) (➔ Page 103 *Windowing tools*)
 Top right corner menu	Measurement and marking tools (➔ Page 109 *Measurement tools*) (➔ Page 108 *Annotation tools*)
 Bottom right corner menu	Window settings and modality-specific presets (➔ Page 103 *Windowing tools*)

Bottom left corner menu	Visualization functions and display types (➔ Page 111 *Available display types*)

5.4.4 Information on images

For orientation and better access to the images, various information is displayed on the images.

If relevant image areas are covered, you can hide the image text or the graphical objects on the images. Click the **Full Text** icon, the **Custom Text** icon, or the **Hide Graphics** icon in the **Favorite Tools** panel.

Image text

Relevant information from the DICOM image header is displayed as text in each corner.	

Image quality

Information on the image quality is displayed in the image text of the lower left corner of a segment:

- **CQ** (compression quality)

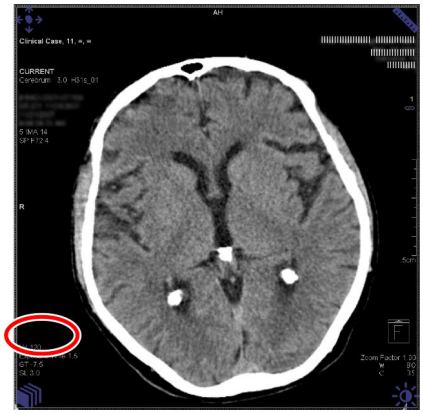
  Indicates the image compression ratio in percent. If the compression ratio is unknown, **CQ!** is displayed.

- **TC** (transfer compression)

  Indicates that the image information is not complete, as transfer from server to client has not been finished.

- **LQR** (low quality rendering)

  Indicates that the displayed image is rendered in lower quality than possible (displayed only temporarily)



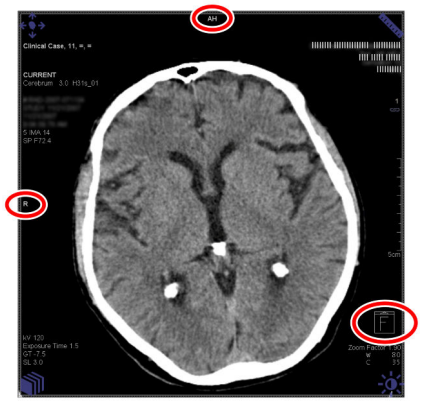
Orientation markers

Orientation Markers are displayed in the upper middle and the middle left of the segment. The following designations are possible:

- **A(nterior), P(osterior)**
- **R(ight), L(eft)**
- **H(ead), F(oot)**
- **X** (unspecified)

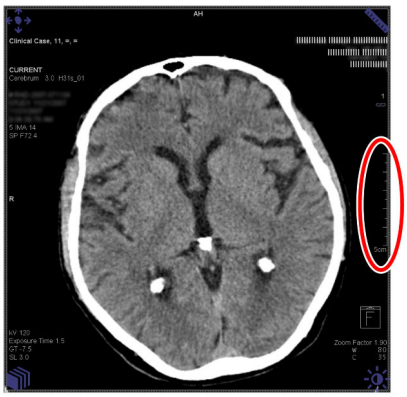
Multiple designations such as **PF/AH** (main orientation first) are also possible.

An orientation cube is displayed. The orientation cube indicates the anatomical orientation.

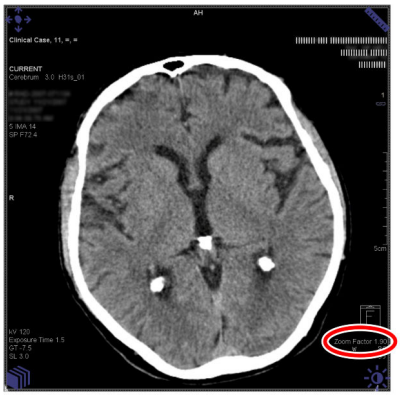


When you flip or rotate the image, the orientation markers change accordingly.

Scale bar

A scale bar is displayed in the right of the segment. You can use this scale bar as a guide. One unit represents 1 cm. The scale bar is only displayed if it does not cover the image text.	

Zoom factor

The current zoom factor is displayed in the image text.	

Window values

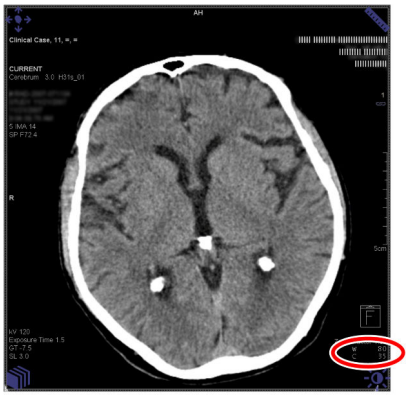
The Windowing values **W(idth)** and **C(enter)** indicate the current values for contrast and brightness. In PET, NM, or SPECT images the window is defined by the position of the lower and upper window limit; **B(ottom)** and **T(op)**.	

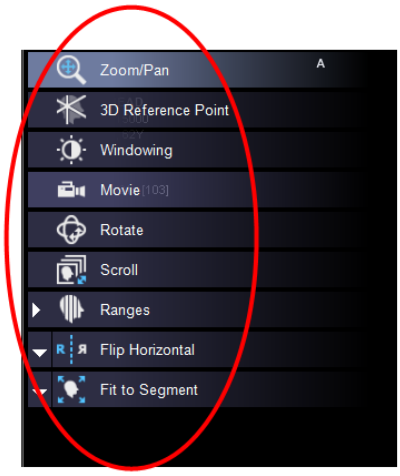
Image number in MPR and MPR/MPR images

In MPR and MPR/MPR images, the slice position of the original (2D) image is shown.	

Corner menus

Corner menus are available in each corner of each segment.

The corner menus allow you to select tools without leaving the segment.

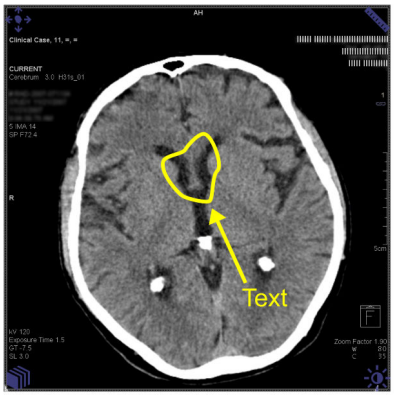


Annotations and graphical objects

Annotations and graphical objects drawn for labeling or measurements are laid over the images and can be saved on a snapshot.

These include:

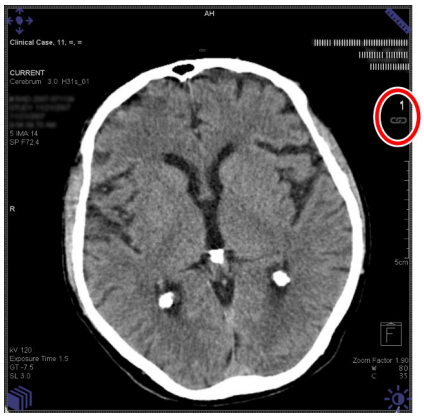
- Annotations
- Markers and arrows
- ROIs and VOIs
- Lines and angles
- Density measurements



Watermarks and mini toolbars

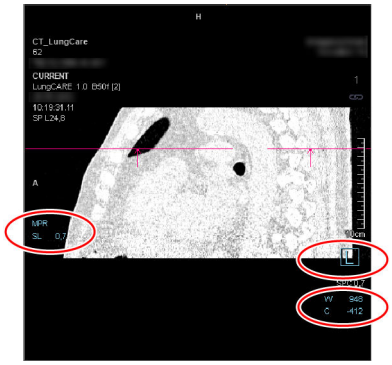
For certain tools, such as **Movie**, a watermark is displayed in the lower middle of the segment. When you move the mouse over the watermark, the corresponding mini toolbar is displayed.	

Synchronization of segments

When the images are synchronized, a chain in the upper right of the segment indicates this. All segments belonging to one group of synchronized segments are marked with the same number. You can synchronize images by selecting a layout that contains synchronization groups. You can also synchronize segments temporarily for specific actions, for example, zoom/pan: press the **Ctrl** key and select the required segments and perform the required action	

5.4.5 How to use interactive image text

With interactive image text you can easily adapt values. Interactive image text is displayed as blue-green image text.



You can interact with interactive image text in different ways:

- Click to switch between predefined values

  This is only possible if for this kind of image text predefined values are existing.

- Drag to increase or decrease the value

  The value changes in small intervals (available only for slice thickness).

  The value changes in large intervals.

- Double-click to enter values with the keyboard

  This is only possible for numeric values.

  Editable values are marked with a frame when hovering the mouse over the value.

- Click the orientation cube to toggle through the standard orientations.
- Double-click the orientation cube to return to the initial position.

5 Getting started

6 Clinical routine with *syngo*.via

Working in *syngo*.via is workflow-oriented. *syngo*.via provides basic workflows that you can use for a wide range of examinations and disease-specific workflows that are especially designed for certain examinations.

Depending on your clinical workflow and installation, you can start a workflow in *syngo*.via or from your RIS or PACS work list. The following procedure is an example of how you can handle workflows in your radiology department.

Note that you need to have the role *Technologist* to initially open an assigned workflow. (➔ Page 17 *User profiles*)

1. A technologist performs an examination at a modality.

   The acquired images are sent to *syngo*.via.

2. *syngo*.via assigns a certain workflow to the images, performs preprocessing, and provides the workflow in a work list.

   You can assign a workflow manually if no workflow has been mapped. (➔ Page 79 *Assigning workflows to studies*).

   You can load data in different ways from the Patient Browser into the Viewer. (➔ Page 81 *Loading patient data for reading*).

3. If necessary, a technologist optimizes and prepares the display of images and saves the workflow for a radiologist. (➔ Page 127 *Handing over a workflow*)

4. A radiology resident opens the workflow, starts to read the images and creates findings.

5. To review the findings later on with an attending radiologist, he saves the workflow. (➔ Page 127 *Saving a workflow*)

Data is closed but all measurements, annotations, findings, and other changes are retained. The status of the workflow in the work list is set to *Saved*.

6 The attending radiologist opens the procedure, reviews the findings, and adds another finding.

7 The attending radiologist completes the workflow and the report data is sent to the connected RIS. (→ Page 129 *Saving and archiving a workflow*)

Data is closed, the workflow is completed, the results are sent to the configured archive.

6.1 Patient Browser

The **Patient Browser** is the entry point to find and manage studies and to start the reading workflow.

(→ Page 69 *Screen layout of the Patient Browser*)

(→ Page 71 *Opening the Patient Browser*)

In the **Patient Browser** you can do the following actions:

- Search for data, and sort and refresh the search results

  (→ Page 73 *Searching for studies in the Patient Browser*)
- Create work lists and customize them
- Load data for reading

  (→ *Loading of patient data*)
- Manually assign workflows to studies

  (→ Page 78 *Workflow assignment*)
- Group and ungroup studies for combined loading
- Import and export data

  (→ Page 132 *About the export of data (images)*)
- Correct patient data as well as study, series, or instance data

  Rearrange studies, series, or instances

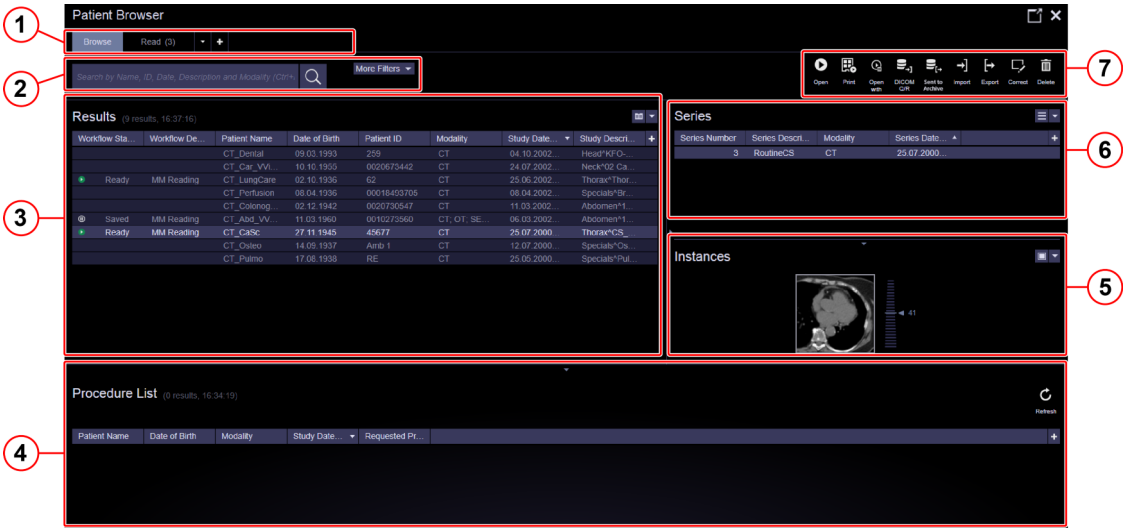
- Delete studies, series, or instances from the Short Term Storage (STS)
- Select data for archiving

  (➔ Page 143 *Selecting data for archiving*)
- Search for prior or related studies on external media or DICOM nodes and import these studies to your local system
- Display DICOM header attributes

You can configure the **Patient Browser** in the ***syngo*.via Administration Portal** and the **Configuration Panel**.
(➔ *Configuration of the Patient Browser*)

6.1.1 Screen layout of the Patient Browser

The **Patient Browser** consists of the following areas:



(1) **Work list** bar

Shows up to five work lists

Enables you to create lists

(2) **Search** area

Enables you to set the current search criteria to search for data or to set up work lists

(3) **Results** list

Shows the data that matches the current search criteria in two different views:

- **Studies**: Lists the studies one by one

- **Patients**: Lists the studies grouped on patient level

(4) **Procedure List**

Enables searching for scheduled procedures and assigning studies and series to a selected scheduled procedure

This list is not available for all user roles.

(5) **Instances** list

Shows the images and results of the selected series in three different views:

- **List**: Lists the instances of the selected series

- **Preview**: Shows a preview of the selected series. You can scroll through the preview.

- **Thumbnails**: Shows a thumbnail for each instance of the selected series

(6) **Series** list

Shows the series of the selected study in two different views:

- **List**: Lists the series of the selected study

- **Thumbnails**: Shows a thumbnail for each series of the selected study

(7) **Toolbar**

(→ Page 71 *The toolbar of the Patient Browser*)

(→ *Toolbar of the Patient Browser*)

Depending on your screen orientation (landscape or portrait), the layout of the **Patient Browser** may vary.

6.1.2 Opening the Patient Browser

If the **Patient Browser** is not visible due to certain actions, for example, you closed the **Patient Browser**, or opened the **Configuration Panel**, you can open it again.

- On the access bar, click the **Patient Browser** icon to display the **Patient Browser** again.

6.1.3 The toolbar of the Patient Browser

The toolbar of the **Patient Browser** provides access to the following tools.

Open	Loads patient data for reading with the assigned workflow. If no workflow is assigned, a default workflow is assigned and the workflow is started. The corresponding Patient tab is displayed on the access bar. Keyboard shortcut: **Ctrl** + **O** (→ Page 81 *Loading patient data for reading*)
Open with	Lets you select the workflow and loads patient data for reading The corresponding Patient tab is displayed on the access bar. (→ Page 81 *Loading patient data for reading*)
Print	Opens the **Print** workflow step for printing the selected data Keyboard shortcut: **Ctrl** + **P**
DICOM Q/R	Searches and retrieves studies from external media or DICOM nodes. This function only works if remote DICOM nodes are configured or a CD/DVD/USB media containing a DICOMDIR is inserted on the local client PC. Keyboard shortcut: **Ctrl** + **R**
Send to Archive	Sends the selected series to the default archive without delay This icon is only available, if a default archive is set. Keyboard shortcut: **Ctrl** + **Shift** + **S**

Import	Imports files from external media, file systems, or network nodes This function works even if no DICOMDIR is available. Keyboard shortcut: **Ctrl** + **I**
Export	Exports DICOM data to external media, file systems, or network nodes Keyboard shortcut: **Ctrl** + **E** (→ Page 132 *About the export of data (images)*)
Correct	Lets you correct patient data
Delete	Deletes data from Short Term Storage (STS) Only available if the data is not protected from deletion. Note that the data cannot be restored after deletion.

6.1.4 Searching for studies in the Patient Browser

In the **Patient Browser**, you have different options to find studies:

- Using the **Quick search** field
- Using the **More Filters** area
- Using work lists or demo lists (→ Page 76 *About work lists*)

1 In the **Quick search** field, enter your search criteria:

- Patient Name
- Date of Birth
- Patient ID
- Study Description
- Study Date
- Accession Number
- Modality

Ensure you adhere to the DICOM format (YYYYMMDD) or to the Windows regional settings for date and time formats.

(→ Page 75 *Search syntax and rules in the Patient Browser*)

2 To refine your search, click **More Filters** (or a criterion) and enter additional search criteria in the search condition fields of the **More Filters** area.

If required, you can add or remove search condition fields.

Note that the entry in the **Quick search** field is always part of your query.

3 Click the **Search** icon.

While the search process is running, the **Search** icon changes its shape. To abort the search process, click this icon.

The search results are displayed in the **Results** list.

The **Results** list is updated periodically.

The **Results** list is limited to 1000 matches.

Search syntax and rules in the Patient Browser

The **Patient Browser** search supports common basic search syntax and rules.

(➔ Page 73 *Searching for studies in the Patient Browser*)

Syntax and rules for names and values:

- Text strings are **not** case-sensitive, for example, **`Meyer`** and **`meyer`** lead to the same results.
- You can enter a fragment of a name or a value, for example, **`Me`** to search for all patients whose name begins with "Me".
- You can use **`*`** as a wildcard character.

  Examples:

  – **`*eyer`** will find "Meyer", "Breyer", "Dreyermann", etc.

  – **`Me*er`** will find "Meyer", "Meier", "Meuller", etc.
- You can search for patients with composite surnames.

  In such cases, you have to enter the surname in the appropriate order in the **Quick search** field. If you just enter parts of the surname, the patient will not be found.

  Example: The last name "van Houten" must be entered starting with **`van Hou`**.
- In the **More Filters** area, you can use quotation marks "" to search for an exact term.

  Example: "**`Me`**" will only find "Me" but not "Meyer", "Meier", "Merz", etc.

  If you do not want to search for an exact term, you can also use quotation marks "" combined with **`*`** as a wildcard character.

  Example: "**`van Hou*`**" will find "van Houten"

Syntax and rules for date and time ranges:

- In the **Quick search** field, you can enter a date manually.
- You can combine a date or a time range with a name or a value.

  Example:

  `2013/10/25 Meyer` will find all patients whose name is "Meyer", and whose birth date is "2013/10/25".

Ensure you adhere to the DICOM format (YYYYMMDD) or to the Windows regional settings for date and time formats.

6.1.5 About work lists

In the **Patient Browser**, work lists are predefined queries to find studies. Additionally, you can create demo lists (for example, for clinical rounds, discussions) by adding selected studies to it. The figure next to the work list name indicates the number of matching items.

(→ Page 77 *Creating a work list for the current filters*)

Up to five work lists are shown on the work list bar. Additional work lists are available when clicking the **More Work Lists** icon.

By using drag and drop, you can arrange the order of work lists and move them to or from the **More Work Lists** menu (hide/show).

Work lists basing on filters

The work lists for your daily work are dynamic, for example, a work list shows all series that are ready to read. These work lists are based on filters. Studies are automatically added or removed from the work list due to their filter settings.

Work lists basing on collected studies (demo list)

Demo lists are manually created work lists. You just add the required studies to such work lists. These work lists are designed for presentation, conference, discussion, or training purposes.

As long as a study is part of a demo list it will not be deleted from the short term storage (STS) by the automatic data deletion procedure, although it can be deleted manually.

In the **More Filters** area, administrators can search for studies that are part of a demo list. Information whether the study is part of a demo list can be shown in the **Demo List** column of the **Results** list.

Public work lists

Public work lists are customized work lists that are available for all users. To create and modify public work lists, you need clinical administrator rights.

Customized work lists

You can create, modify, and delete user-specific work lists according to your needs.

Purpose	Settings in the Search area
Reading everything	**Workitem Type**: Read **Workflow Status**: Ready or Saved
Reading everything from a certain modality type	**Workitem Type**: Read **Workflow Status**: Ready or Saved **Modality**: for example, **MR**
Reading everything where study description matches a pattern	**Workitem Type**: Read **Workflow Status**: Ready or Saved **Study Description**: for example, **MR*Head***
Checking for all available tasks (for administrative purposes)	**Workflow Status**: Ready or Saved (Use this work list to check for open tasks throughout all work lists at the end of a day.)

Example work lists for various purposes

Creating a work list for the current filters

In the **Patient Browser**, you can customize and save work lists. As a clinical administrator, you can save work lists for all users (public work lists).

1 To clear the entry in the **Quick search** field, click the **X** in the **Quick search** field.

Note that any entry in the **Quick search** field will be part of your work list.

It is recommended to clear the **Quick search** field before creating a new work list.

2 Click **More Filters** and enter your search criteria in the search condition fields of the **More Filters** area.

To avoid outdated filter criteria, use relative date and time formats such as *Today*, *Yesterday*, *Last 2 Days*.

3 If required, you can add or remove search condition fields.

4 In the **More Filters** area, click the **Save Work List** icon.

The **Save Work List** dialog box opens.

– or –

On the work list bar, click the **Create new Work List** icon.

The **Save Work List** dialog box opens.

5 In the **Save Work List** dialog box, select **Create a Work List using the current filters**, to create a work list that uses the current filter settings.

6 To save the work list for all users, select the **Available for all users** check box.

This option is only available for clinical administrators.

7 Enter the name of the work list and click **OK**.

The work list is saved and displayed on the work list bar.

6.1.6 Workflow assignment

The system automatically assigns workflows to studies based on configured rules.

In the **Patient Browser**, you can check the assignment in the **Workflow Description** or **Workflow** column of the **Results** list. When the column is empty, no workflow has been assigned.

- When you double-click a **study** that has no workflow assigned, either the most suitable or the default workflow is assigned and the reading is started.
- When you double-click a **series** or an **instance** of a study that has no workflow assigned, the most appropriate workflow is started with limited write access.

You may also be interested in the following:

- (➔ Page 79 *Assigning workflows to studies*)
- (➔ Page 80 *Batch assigning workflows to studies*)
- (➔ *Workflow assignment on procedure level*)
- (➔ *Workflow assignment on image level*)

Assigning workflows to studies

In the **Patient Browser**, you can manually assign workflows to studies according to your needs.

(➔ Page 78 *Workflow assignment*)

✓ The **Patient Browser** is open.

1 In the **Results** list of the **Patient Browser**, select the study.

When you select and assign multiple studies, all studies belong to the same workflow.

- The most recent data is loaded into the current time point.
- Older data is loaded into prior time points.

2 Right-click and choose **Assign Workflow** from the context menu.

– or –

Click the **Open with** icon.

3 From the list of workflows, select the appropriate workflow.

You can mark frequently used workflows as favorites by clicking the asterisk next to the workflow name.

The new workflow is assigned to the study.

Batch assigning workflows to studies

In the **Patient Browser**, you can assign the same type of workflow to all selected studies in the result list without combining them to a single workflow.

(→ Page 78 *Workflow assignment*)

✓ The **Patient Browser** is open.

✓ You have clinical administrator rights.

1 In the **Results** list of the **Patient Browser**, select multiple studies.

2 Right-click and choose **Batch Assign Workflow** from the context menu.

3 From the list of workflows, select the appropriate workflow.

The same type of workflow is assigned to the selected studies. Nevertheless, every study has its own workflow.

Switching the currently active workflow

You can switch the currently active workflow if it does not fit your needs.

If you want to perform specific image evaluation not possible with the currently active workflow, you can add an additional workflow step instead of switching the workflow.

✓ A workflow has been started and images are loaded for reading.

1 In the workflow step, click the **Switch Workflow** icon.

2 From the list of workflows, select the appropriate workflow.

The current workflow is closed and unsaved results are saved.

The new workflow is assigned to the study.

The Patient tab is opened, the workflow is started, and the images are loaded for reading.

6.1.7 Loading of patient data

In the **Patient Browser**, you can load patient data for reading.

You can simultaneously load data for different patients which are then represented by different Patient tabs.

CAUTION

Patient mix-up is possible if data from different patients are open in different applications.

Diagnosis and treatment may be based on incorrect information.

- Check the patient name in each application.

(➔ Page 81 *Loading patient data for reading*)

(➔ Page 82 *Loading additional data from the Patient Browser*)

Loading patient data for reading

In the **Patient Browser**, you can load studies, series, or instances for reading.

CAUTION

Additional images may arrive at the system while reading a patient.

Diagnosis may be made with incomplete data.

- Check the list/thumbnail view in the **Series** panel for additional images before completing the diagnosis.

1 In the **Patient Browser**, select the corresponding study, series, or instance that should be opened.

Studies that are marked by the "runner" icon in the **Workflow Status** column are preloaded and therefore open quicker.

2 To open the selected element with the assigned workflow, click the **Open** icon.

If no workflow is assigned, a default workflow is assigned and the workflow is started.

When you open a workflow that is opened by another user the workflow is opened with limited write access.

– or –

To assign another workflow before opening, click the **Open with** icon and select the appropriate workflow.

The Patient tab is opened, the workflow is started, and the images are loaded for reading.

If you re-open a workflow, the workflow will be displayed in the previous view. Ensure that the segments are displaying the data you expect.

If the image area remains empty after loading:

- The workflow assignment may not match. Use the **Series** panel to drop data into the image area.
- The default layout may not match. Select another layout.

Loading additional data from the Patient Browser

In the **Patient Browser**, you can load additional studies, series, or images for an already loaded patient.

CAUTION

The active patient could contain data from other patients or series (for example, after user interactions, data import from CD/DVD, or merge from RIS).

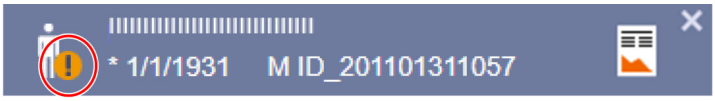
The diagnosis and treatment may be based on incorrect information.

- Check that the data refer to the active patient before using the data.

1. In the **Patient Browser**, select the desired data from the **Results** list, **Series** list, or **Instances** list.
2. Make sure that the selected data really belongs to the same patient.
3. Drag the data onto the corresponding Patient tab.

   If additionally loaded data is not identified as belonging to the same patient, confirm the load operation.

   In this case, the Patient tab displays an indicator for inconsistencies.



Depending on the layout, data is displayed in the image area.

(→ *See: Adding studies to the Series panel*)

6.2 Viewer

In the *syngo*.via Viewer, you can manipulate images, create and review findings, and print images.

Display of data after loading

Depending on the workflow that has been assigned, images or series are pre-processed, loaded, and are displayed in an appropriate layout.

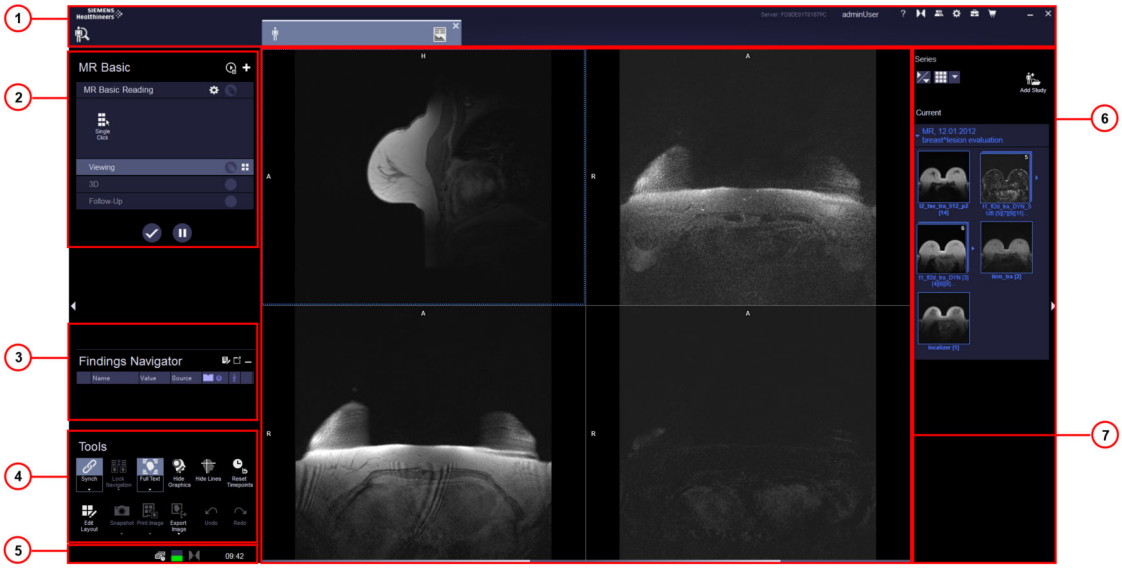
If the image area remains empty after loading:

- The workflow assignment may not match. Use the **Series** panel to drop data into the image area.
- The default layout may not match. Select another layout.

6.2.1 Screen layout of the Viewer

The layout of the Viewer depends on the selected workflow.

After you have loaded the patient data into an advanced workflow, the screen shows the following areas:



(1) **Access Bar**

The access bar enables you to switch between the **Patient Browser** and the Patient tabs.

Additionally, the access bar displays your user name, the currently selected server, and provides access to the **Configuration Panel**, Online Help, and other services.

(2) **Workflow steps** area

Displays the current workflow steps and corresponding functions.

Below the workflow steps, workflow controls allow you to complete and save workflows.

(3) **Findings Navigator**

The **Findings Navigator** allows you to navigate through evidence objects (measurements, graphical objects, annotations, and snapshots) that are declared as findings.

(4) Common **Tools** area

The Common **Tools** area provides the common functions.

(5) **Status** bar

The status bar is an area at the bottom of the screen that displays information about the system status and the current time.

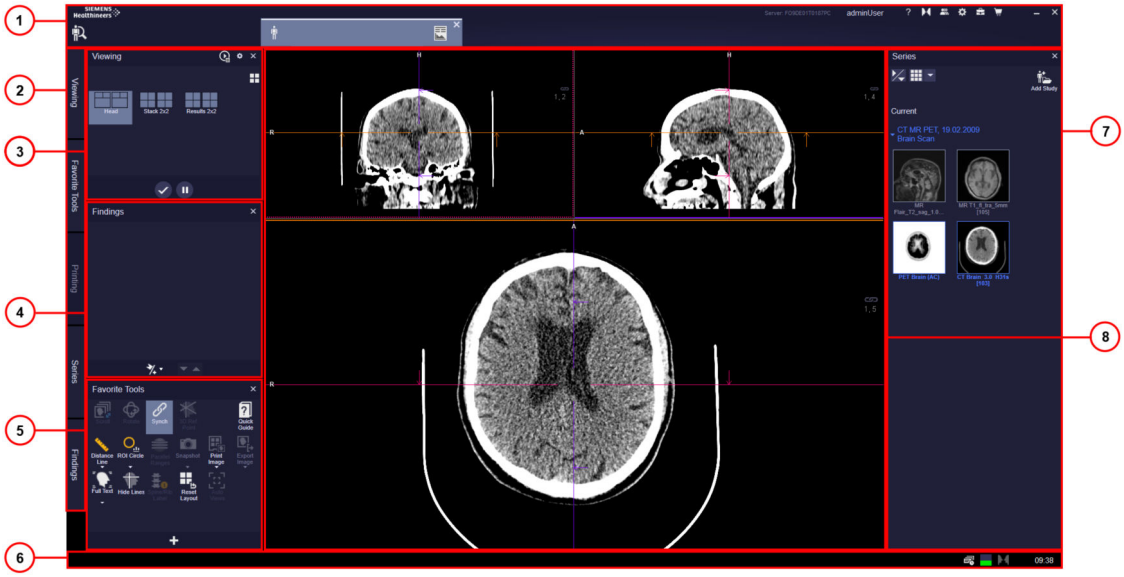
(6) **Image** area

The image area displays images and series in segments. Corner menus are available in the segments.

(7) **Series** panel

The **Series** panel lists all currently loaded data within a workflow. You can load data groups, functions, as well as switch time points in the image area.

When you load the patient data into MM Reading, the screen shows the following areas:



(1) **Access Bar**

The access bar enables you to switch between the **Patient Browser** and the Patient tabs.

Additionally, the access bar displays your user name, the currently selected server, and provides access to the **Configuration Panel**, Online Help, and other services.

(2) **Tabs**

Indicate the individual parts of the viewer. Clicking a tab opens a panel.

(3) **Viewing** panel

Displays the preferred layouts and provides access to the **Layout Gallery**.

On the bottom of this panel you find the controls to complete and save the workflows.

(4) **Findings** panel

Allows you to navigate through evidence objects (measurements, graphical objects, annotations, and snapshots) that are declared as findings.

In the **Findings** panel, you can also group, classify and comment the created findings.

(5) **Favorite Tools** panel

Provides your favorite functions.

The **Favorite Tools** panel is configurable. From here, you can open the **Tool Gallery** to choose the tools that should be displayed on that panel.

(6) **Status** bar

The status bar is an area at the bottom of the screen that displays information about the system status and the current time.

(7) **Series** panel

The **Series** panel lists all currently loaded data within a workflow. You can load data groups, functions, as well as switch time points in the image area.

(8) **Image** area

The image area displays images and series in segments. Corner menus are available in the segments.

6.2.2 MM Reading workflow

By default, *syngo*.via provides the basic workflow **MM Reading**, which is used to view and compare images and series of any modality, including 2D, 3D, and 4D data sets.

Typically, **MM Reading** is opened as a separate workflow, but it can also be integrated into other workflows as a workflow step, where it can be used to read results from other workflow steps.

Depending on whether you have **MM Reading** as a separate workflow or as an additional workflow step, the layout and the available tools differ.

MM Reading as workflow

If you are using **MM Reading** as a workflow, the screen layout consists of an image area and several panels.

- **Viewing** panel: provides the preferred layouts as well as access to the **Layout Gallery**, and the controls to complete and save the workflow.
- **Favorite Tools** panel: provides your preferred tools for viewing images and also access to the **Tool Gallery**, which includes all available viewing tools. The **Favorite Tools** panel is configurable.

  (➔ Page 100 *Tool Gallery*)

  (➔ Page 101 *Adding tools to the Favorite Tools panel*)
- **Findings** panel: allows you to create structured findings according to dedicated finding types. Furthermore, you can group, classify, and comment created findings, and add body regions to assign the created findings to these body regions.

  (➔ Page 125 *Findings panel*)
- **Series** panel: provides an overview of the series and instances that belong to the loaded study.

  (➔ Page 96 *Series panel*)

The panels can be resized and rearranged on the screen. You can either drag them to another predefined position on the screen or make them a floating window.

You can also hide the panels to enlarge the image area. The panels are then displayed as tabs along the side of the screen. Clicking these tabs displays the panels again at their previous position.

In addition, you can create a report by using the **Report Editor**. See (➔ *Report Editor*) in the *Reporting Online Help*

MM Reading as workflow step

If you have integrated the **MM Reading** workflow in another workflow as a workflow step, for example, CT Vascular, the screen layout and the available tools differ. (→ Page 84 *Screen layout of the Viewer*)

The flexible and configurable layout is not available in this case.

All measurements marked as findings are collected in the **Findings Navigator** instead of the **Findings** panel. (→ Page 127 *Findings Navigator*)

The **MM Reading** workflow step also provides the set of preferred layouts. You can change the preferred layouts in the **Assign Layouts** dialog box, which is accessible in the **Configuration** dialog box of the **MM Reading** workflow step.

The configuration of preferred layouts must only be done once. This configuration takes effect in all forms of appearances of **MM Reading** (workflow or workflow step).

6.2.3 Status icons in workflow steps

In the Viewer, small icons next to the name of the workflow step indicate the status:

	The workflow step is available in the workflow, but no data is currently available.
	The workflow step is ready and can be processed. Data is available.
	The workflow step has been activated. The icon is displayed when you click on the workflow step the first time.

	The workflow step is completed.
	An error occurred during the workflow step.
	Information on the workflow step is available. Click this icon to display the system messages. After reading and confirming the system messages you can continue working.

6.2.4 Layout Gallery

A layout determines the number, size, and order of segments, as well as the data group displayed in segments. *syngo*.via provides a number of predefined layouts.

Click the **Layout Gallery** icon to display all available layouts.

In MM Reading a set of preferred layouts is shown in the **Viewing** panel.

Depending on the workflow, the style and functional range of the **Layout Gallery** may differ.

- (➔ Page 92 *Screen layout of the Layout Gallery of advanced workflows*)

For each workflow, specific layouts are defined. The layout that is applied when you open data for processing or viewing depends on the active workflow step.

You can select another layout if the applied layout does not fit your needs or if the segments are empty. The default layout, which is defined for the active workflow step, is applied when you open the same type of images with the same workflow.

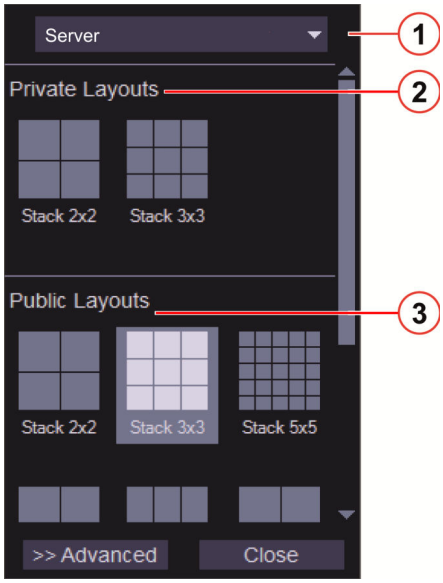
In addition, **MM Reading** provides the **Segment Configuration** for advanced layout definition. With this functionality you can configure alternative series for the image segments to be displayed if the original series is not available.

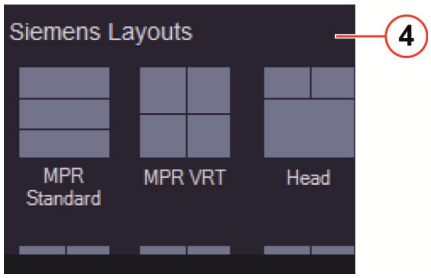
Screen layout of the Layout Gallery of advanced workflows

In each *syngo*.via workflow, predefined layouts are available.

In the **Layout Gallery**, layouts are arranged in collections. Depending on your requirements for the current evaluation, you can change the displayed layout by selecting another one from the collection.

Depending on your user rights, you can adapt the collections to your needs, for example, by dragging layouts from the **Public Layouts** section to the **Private Layouts** section.





The visible layout sections (Siemens Healthineers, public, private) depend on your user role

(1) **List of users** (only visible for clinical administrators)

The clinical administrator can select the private layouts from any user of the site. The clinical administrator can drag these layouts to the public layouts.

The name of the user is defined as follows:

`<Servername>_<Login name>`.

(2) **Private Layouts**

Collection of your own personal layouts. You can create, modify, and delete your own private layouts.

For clinical administrators, this section contains the private layouts of the user selected in the List of users.

Note: If you have no **Private Layouts** section, you are not allowed to create private layouts. For more information, please contact your clinical administrator.

(3) **Public Layouts**

Collection of layouts that have been predefined by a clinical administrator for your entire institution.

Only administrators are allowed to edit public layouts.

(4) **Siemens Layouts** (only visible for administrators)

Collection of layouts provided by Siemens Healthineers. These layouts may be updated when the system is updated or upgraded.

Note that the layouts provided by Siemens Healthineers cannot be deleted or modified.

When you save a workflow, the active layout is sent along with the data.

When your colleague starts working with the data, this layout becomes temporarily available. The layout name is put in parentheses, for example *(Stack 2x2)*.

When a workflow has been re-opened by you, the layout is selected by default.

Layout tools

Depending on the workflow, you have several tools available to change the displayed layout, modify the existing layouts or to create a new layout.

Layout Gallery	Opens the Layout Gallery Here you can select a layout, create a new layout, or save changes to the current layout.

In the Layout Gallery, click the **>>Advanced** button to access more functions. In MM Reading, you can find more layout functions in the **Layout Gallery** above the set of predefined layouts.	
New	Opens the **Layout Creator** Here you can define the number of monitors and segments for a new layout. This option is not available in MM Reading.
Edit	Activates the editing mode for layouts In this mode, you can manipulate the size of the currently displayed segments. Keyboard shortcut: **Alt** + **L** This option is not available in MM Reading.
Save	Saves changes in the selected layout
Save As	Saves the selected layout as a new layout

Additional functions, for example, for renaming a layout, are available in the context menu of a layout icon in the **Layout Gallery**.

In Favorite Tools panel/Common Tools area	
Reset Time Points	Resets time points which you have applied from the **Series** panel to the time points that are defined in the layout (→ *Switching and resetting time points*)

Edit Layout	Activates the editing mode for layouts In this mode, you can manipulate the size of the currently displayed segments. Keyboard shortcut: **Alt** + **L** This option is not available in MM Reading.

6.2.5 Series panel

The **Series** panel gives an overview of the series of the loaded study and groups this data according to specific criteria.

The **Series** panel provides two views:

- **List** view: gives an overview of the data sorted into data groups
- **Thumbnails** view: gives a preview of the images

Depending on the layout, not all available data might be displayed in the image area. Data that is displayed in the image area is indicated by bold type in the **Series** panel. You can use the **Series** panel to select further images to be displayed in the image area.

The **Series** panel also offers you an option to add further studies to the currently loaded one, for example, prior studies.

(→ Page 96 *Screen layout of the Series panel*)

(→ Page 98 *Adding studies to the Series panel*)

Screen layout of the Series panel

The **Series** panel is always available in the Viewer application. You can hide or show it by using the small arrow head icon on the right side of the screen.

(1) **Expand and collapse hierarchy** icon

(2) **Change view** icon

Gives an overview of the data sorted into data groups

List view: gives a textual presentation of the data groups

Thumbnails view: gives a preview of the data groups

(3) **Images** listed according to data attributes

The sorting is structured hierarchically and the different levels are usually sorted by image attributes, for example, modality and study date on first level, and series date and time on second level.

(4) Result icon

Indicator that a result series has been created

(5) Expand or collapse **Series** panel

(6) **Add Study** icon

Opens the **Add Study** dialog box for loading additional patient data (→ Page 98 *Adding studies to the Series panel*)

(7) **Expand or collapse series stack**

Dynamic series are displayed as stacks in the **Thumbnails** view and can be expanded with the triangle icon to show all series in the stack.

If you have selected the **List** view in the **Series** panel, you can change the font size by pressing the **Ctrl** key and using the scroll wheel.

Adding studies to the Series panel

In the **Series** panel, you can load additional studies for an already loaded patient. You can search for prior or related studies on DICOM nodes or external media.

syngo.via suggests a list of data belonging to the already loaded patient.

✓ You have appropriate user rights to change sources and filters.

1 In the **Series** panel, click the **Add Study** icon.

The **Add Study** dialog box opens.

The **Source Selection** and the **Search Filters** are preset and can be changed by users with appropriate rights.

The preset sources are the prefetch nodes and the default archive.

2 Click the **Search** icon.

The results are listed.

3 Select a study or series and click the **Load Data** icon to load the data into the **Series** panel.

The column **Data Source** may display one of the following values:

- **Local**:

  Data is located on the short term storage. It can be loaded immediately.

- Name of a PACS, for example, **MY_FAVORITE_PACS**:

  Data is located on another network node. Loading it requires retrieving and may take some time.

Data can also be loaded from removable media to the client machine (for example, from CD/DVD or USB stick).

Depending on the layout, data is displayed in the image area. All data is available in the **Series** panel.

Adding and replacing series in the image area

From the **Series** panel, you can add and replace series in the image area.

You can drag a series to an empty segment to add the series to the display. You can also drag a series to a segment that already contains a series to replace the currently displayed one.

The images can be dragged from the **List** view or from the **Thumbnails** view.

1. In the **Series** panel, select the series you want to be displayed. (Multiple selection with the **Ctrl** key.)
2. Drag the series into a segment.

   The selected image is displayed in the required segment.

   – or –

   Keep the **Ctrl** key pressed and drag the series to the image area.

   The series are added to the already displayed series and the layout is adapted accordingly.

   – or –

   Right-click and drag the series into a segment.

   The mouse pointer changes shape and a context menu is displayed.

3 From the context menu, choose **Insert before** or **Replace**.

In the image area, the series are displayed accordingly.

6.2.6 Basic functions to read images

This is your reference of functions and tools in the corner menus and in the **Favorite Tools** panel. Different tools are displayed in the corner menus depending on the loaded images. Additional tools are located in the workflow steps area depending on the workflow.

For the U.S.A., please note the following:

- Monitors (displays) should not be used for diagnosis unless the used monitor (printer) has specifically received 510(k) clearance for this purpose.
- Paper printouts should not be used for diagnosis unless the used Post Script printer has specifically received 510(k) clearance for this purpose

Tool Gallery

The **Tool Gallery** is only available in MM Reading.

The **Tool Gallery** provides all available tools, sorted according to their category, for example, **Ranges**, **Measurements**, or **Display Types**.

You can use the **Tool Gallery** to add frequently used tools to the **Favorite Tools** panel. Thus, your favorite tools are always available. (→ Page 101 *Adding tools to the Favorite Tools panel*)

Another option is to open the tools directly from the **Tool Gallery**. Therefore, you can leave the **Tool Gallery** permanently open, or you can open it every time you want to use a specific tool. (→ Page 101 *Calling functions from the Tool Gallery*)

Adding tools to the Favorite Tools panel

The **Favorite Tools** including configuration and access to the **Tool Gallery** are only available in MM Reading.

You can add frequently used tools from the **Tool Gallery** to the **Favorite Tools** panel of the **Viewer**. You can also remove tools from this panel if you do not need them anymore.

1 In the **Favorite Tools** panel, click the **More** icon to open the **Tool Gallery**.

2 In the **Tool Gallery**, select an icon and drag it to the **Favorite Tools** panel.

– or –

Right-click an icon and choose **Add to Favorites** from the context menu.

If the menu entry is dimmed, the tool has already been added to **Favorite Tools** panel.

The selected tool is now also displayed in the **Favorite Tools** panel.

To remove a tool from the **Favorite Tools** panel, you can right-click the icon and choose **Remove from Favorites** from the context menu.

You can also open tools from the **Tool Gallery**.
(→ Page 101 *Calling functions from the Tool Gallery*)

Calling functions from the Tool Gallery

You do not need to add a tool to the **Favorite Tools** panel to activate the function. You can activate the required tool directly from the **Tool Gallery**.

Some tools in the **Tool Gallery** are dimmed. This might have the following causes:

- No image segment is selected
- The currently active step or the loaded data set does not meet the requirements for using the tool

1 Select one or more image segments.

2 In the **Favorite Tools** panel, click the **More** icon to open the **Tool Gallery**.

3 Click the required tool icon.

The selected function or mode is applied.

Scrolling tools

You can use the mouse or the keyboard to scroll through images. When you use the scroll wheel or the arrow keys, the images are scrolled one after the other without skipping images. When you scroll with the right mouse button it can happen that some images are skipped. If this mode is activated, the string "Skip" is displayed in the segment. You can enable scrolling without skipping (**U** key).

In stripe display mode or in striped stack display mode, a scrollbar is provided.

Top left corner menu	
Scroll	Activates the left mouse button for scrolling **Top left corner menu** > **Scroll** Keyboard shortcut: **I**
Movie	Scrolls automatically through series, 3D volumes, or data groups **Top left corner menu** > **Movie** Keyboard shortcut: **Spacebar**

Tool Gallery	
Next Study / Previous Study	2D Reading Switches to later or earlier study All relevant studies need to be displayed in the **Series** panel of MM Reading

Windowing tools

You can modify the brightness and contrast of an image. Windowing often helps you to recognize suspicious anatomical structures on images.

Top left corner menu	
Windowing	Changes width and center values Note: Windowing is always available by pressing and dragging the middle mouse button (scroll wheel). In the images of most modalities the window is defined by width and center. The displayed windowing values can be changed interactively in the images by double-clicking it and enter a value. In PET, NM, or SPECT images the window is defined by the position of the lower and upper window limit (bottom and top). **Top left corner menu** > **Windowing** Keyboard shortcut: **W**
Bottom right corner menu	
Reset Windowing	Resets the window values (width and center) to the values the image was displayed with after initial loading **Bottom right corner menu** > **Reset Windowing**

Color LUT	Lets you apply color LUTs (look-up tables) from a list For images from different modalities, corresponding default color LUTs are available. **Bottom right corner menu** > **Color LUT**
Presets	Lets you apply windowing preset values from a list **Bottom right corner menu** > **Presets**

Synchronization tools

You have tools to create synchronization groups for comparing series.

Favorite Tools panel	
Synch	Groups selected segments so that you can work with them synchronously For the **Zoom** and **Navigation** options note the following: Depending on the workflow, data sets need to be in the same frame of reference or there is a spatial alignment of the images (for example for images from different time points). During synchronization the following indicators are displayed in the segments: : Synchronization successful : Synchronization not possible
Align	Allows you to adjust the spatial alignment within synchronized segments Keyboard shortcuts: **Ctrl** + **Shift** + **A** to activate the tool **Ctrl** + **Shift** + scrolling to apply the tool

Align Timepoints	Allows you to adjust the spatial alignment within synchronized segments of different time points Keyboard shortcuts: **Ctrl** + **Shift** + **A** to activate the tool **Ctrl** + **Shift** + scrolling to apply the tool Only the registration between time points is modified. The registration within each time point is not affected.
Top left corner menu	
Correlated Cursors	Allows you to activate and deactivate the correlated cursor mode. If activated, a synchronized cursor is displayed in all segments. **Top left corner menu** > **Correlated Cursors**

Optimization and preparation tools

You have tools to adapt or enhance the image display.

Optimization *Top left corner menu*	
Rotate	Rotates a 2D image or a 3D volume **Top left corner menu** > **Rotate** **Alt** + right mouse button Keyboard shortcut: **O**
Flip Horizontal	Flips images left or right **Top left corner menu** > **Flip Horizontal**
Flip Vertical	Flips images up or down **Top left corner menu** > **Flip Vertical**

Zoom/Pan *Top left corner menu*	
Zoom/Pan	Zooms and pans continuously with the mouse **Top left corner menu** > **Zoom/Pan** **Alt** + left mouse button Keyboard shortcut: **Z**
Fit to Segment	Resizes the images in the segment so that they are entirely visible in the segment **Top left corner menu** > **Fit to Segment**
Fit to Acquisition Size	Displays the selected images in their original resolution as acquired (acquisition size) Using this resolution, all objects drawn on images, such as ROIs and graphic objects, are displayed in an optimal way. **Top left corner menu** > **Fit to Acquisition Size**
AutoViews *Bottom right corner menu*	
AutoViews	Navigates to a location of interest and gets the views oriented in a clinically meaningful way. **Bottom right corner menu** > **AutoViews**

Home Position *Top left corner menu*	
Home Position	Resets the following viewing parameters to the values that were initially displayed in the segment: • Orientation • Zoom/pan values • Viewing direction • Alignment in the segment (in-plane rotation) • Image position (for slice and slab images) **Top left corner menu** > **Home Position Size**
Direct manipulation	
drag & drop	Switch / rearrange segments You can arrange available series according to your needs. You can drag a series from the **Series** panel to an empty segment to add the series to the display. You can also drag a series to a segment that already contains a series to replace the currently displayed one. (→ Page 99 *Adding and replacing series in the image area*)

Tools for hiding and showing image overlays

Image overlays, for example, graphical objects, annotations, or image text help you processing the images for evaluation. If required, you can show or hide the image overlays.

Top left corner menu	
Shutters On/Off	Shows or hides existing shutters **Top left corner menu** > **Shutters On/Off**

Favorite Tools panel	
Full Text	Toggles between full or customized image text or hides the image text Keyboard shortcut: **T** **Custom Text** can be configured in the **Configuration Panel**. (→ *"Configuring image text" in the Online Help*)
Custom Text	
Hide Graphics	Shows or hides graphical objects Keyboard shortcut: **G**
Hide Lines	Shows or hides reference lines. Keyboard shortcut: **H**

Annotation tools

You have tools to comment your measurements or further conspicuous areas in the image.

Top right corner menu	
Arrow	Draws arrows onto an image and allows text annotations **Top right corner menu** > **Arrow**
Marker	Marks a position onto an image and allows text annotations **Top right corner menu** > **Marker** Keyboard shortcut: **M**
Text	Allows text annotations **Top right corner menu** > **Text**

Measurement tools

You have tools to mark and measure conspicuous areas in the image.

Top right corner menu	
Distance Line	Measures the length of a line Available for 2D, MPR, MPR Thick, MIP Thin, MinIP Thin, and MPR/MPR images With **Distance Line Properties**, you can specify to turn the display of evaluation results on or off. It can be accessed by clicking the settings icon. **Top right corner menu** > **Distance Line** Keyboard shortcut: **D**
Distance Polyline	Measures the lengths of connected lines Available for 2D, MPR, MPR Thick, MIP Thin, MinIP Thin, and MPR/MPR images With **Poly Line Properties**, you can specify to turn the display of evaluation results on or off. It can be accessed by clicking the settings icon. **Top right corner menu** > **Distance Polyline**
Angle	Measures an angle between two lines Available for 2D, MPR, MPR Thick, MIP Thin, MinIP Thin, and MPR/MPR images **Top right corner menu** > **Angle**

 Pixel Lens	Determines the pixel value at a current position of a pixel or a voxel Available for 2D, MPR, MPR Thick, MIP, MIP Thin, MinIP Thin, and MPR/MPR images With **Pixel Lens Properties**, you can specify the size of the pixel lens. It can be accessed by clicking the settings icon. **Top right corner menu** > **Pixel Lens** Keyboard shortcut: **L**
 ROI Circle	Evaluates statistical values for a circular or elliptical region Available for 2D, MPR, MPR Thick, MIP Thin, MinIP Thin, and MPR/MPR images With **ROI Properties**, you can specify to turn the display of evaluation results on or off. It can be accessed by clicking the settings icon. **Top right corner menu** > **ROI Circle**
 ROI Freehand	Evaluates statistical values for a freely drawn region Available for 2D, MPR, MPR Thick, MIP Thin, MinIP Thin, and MPR/MPR images (Fusion) With **ROI Properties**, you can specify to turn the display of evaluation results on or off. It can be accessed by clicking the settings icon. **Top right corner menu** > **ROI Freehand**
 ROI Polygonal	Evaluates statistical values for a polygonal region Available for 2D, MPR, MPR Thick, MIP Thin, MinIP Thin, and MPR/MPR images With **ROI Properties**, you can specify to turn the display of evaluation results on or off. It can be accessed by clicking the settings icon. **Top right corner menu** > **ROI Polygonal**

 VOI Sphere	Evaluates statistical values for a spherical or an ellipsoidal volume Available for MPR, MPR Thick, MIP Thin, MinIP Thin, and MPR/MPR images (fusion) With **VOI Properties**, you can specify to turn the display of evaluation results on or off. It can be accessed by clicking the settings icon. **Top right corner menu** > **VOI Sphere**
 VOI Freehand	Evaluates statistical values for a volume you have defined drawing freehand contours Available for MPR, MPR thick, MIP Thin, or MinIP Thin images With **VOI Properties**, you can specify to turn the display of evaluation results on or off. It can be accessed by clicking the settings icon. **Top right corner menu** > **VOI Freehand**

You can copy and paste measurement objects between images.

Available display types

Depending on the available image data and the requirements for processing the images, you can apply different display types in each of the image segments.

 2D	Displays 2D images **Bottom left corner menu** > **2D** or by clicking the display type given in the image text Keyboard shortcut: **1**
 MPR	Displays MPR images **Bottom left corner menu** > **MPR** or by clicking the display type given in the image text Keyboard shortcut: **2**

 MPR Thick	Displays MPR Thick images With **MPR Thick Properties**, the properties for this display type can be adapted. It can be accessed by clicking the settings icon. **Bottom left corner menu** > **MPR Thick** or by clicking the display type given in the image text Keyboard shortcut: **3**
 MPR/MPR	Displays fused MPR/MPR images With **MPR/MPR Properties**, the properties for this display type can be adapted. It can be accessed by clicking the settings icon. **Bottom left corner menu** > **MPR/MPR**
 DRR	Displays DRR images With **DRR** (Digitally Reconstructed Radiographs), the images are displayed as x-ray like images. This display type is only available in the **MM RT Image Suite** workflow. **Bottom left corner menu** > **DRR**
 MIP	Displays MIP images **Bottom left corner menu** > **MIP** or by clicking the display type given in the image text Keyboard shortcut: **4**
 MIP Thin	Displays MIP Thin images With **MIP Thin Properties**, the properties for this display type can be adapted. It can be accessed by clicking the settings icon. **Bottom left corner menu** > **MIP Thin** or by clicking the display type given in the image text Keyboard shortcut: **5**

 MinIP Thin	Displays MinIP Thin images With **MinIP Thin Properties**, the properties for this display type can be adapted. It can be accessed by clicking the settings icon. **Bottom left corner menu** > **MinIP Thin** Keyboard shortcut: **6**
 VRT	Displays VRT images With **VRT Gallery**, VRT Presets can be selected and modified. It can be accessed by clicking the settings icon. **Bottom left corner menu** > **VRT** or by clicking the display type given in the image text Keyboard shortcut: **7**
 VRT Thin	Displays VRT Thin images With **VRT Gallery**, VRT Presets can be selected and modified and with **VRT Thin Properties**, the properties for this display type can be adapted. They can be accessed by clicking the settings icon. **Bottom left corner menu** > **VRT Thin** or by clicking the display type given in the image text Keyboard shortcut: **8**
 VRT/MIP	Displays fused VRT/MIP images With **VRT/MIP**, you align a full rendered volume with a 2D projection of a series. This display type is only available in the **MM Oncology** workflow. **Bottom left corner menu** > **VRT/MIP**

 Cinematic VRT	Displays a full rendered volume With **VRT Gallery**, VRT Presets can be selected and modified. It can be accessed by clicking the settings icon. **Bottom left corner menu** > **Cinematic VRT** Keyboard shortcut: **Ctrl** + **7**
 VRT Gallery	Opens the **VRT Gallery** dialog box and lets you select and modify VRT presets. Right-click on **VRT/VRT Thin/Cinematic VRT** and choose **VRT Gallery**.

Registration Tools (Fusion)

You have the tools to align fused base series with overlay series.

 Visual alignment	Allows you to manually align fused base series and overlay series to match them as closely as possible. Available when **MPR/MPR** has been activated. **Top left corner menu** > **Visual alignment**
 Automatic Registration	Fused base series and overlay series are aligned automatically. Available when **MPR/MPR** has been activated. **Top left corner menu** > **Automatic Registration**

Orientation tools

You have tools to quickly navigate through the displayed images.

 3D Reference Point	Allows you to navigate quickly to the same coordinates of a patient in different series and views of series To define a pathology as new center within a segment press **ALT** while clicking it. **Top left corner menu** > **3D Reference Point** Keyboard shortcut: **R**
 Lock Reference Lines	Allows you to lock and unlock the 90 degree orientation of reference lines in a orthogonally synchronized segment. **Top left corner menu** > **Lock Reference Lines**

Tools to mask out structures

You have tools to mask out specific structures in a volume.

 Bone Removal Auto	Masks out bone structures An algorithm is automatically applied depending on the body region. Available for VRT and MIP images **Top left corner menu** > **Bone Removal Auto**
 Bone Removal Head	Masks out bone structures The applied algorithm is optimized for head or neck regions. Available for VRT and MIP images **Top left corner menu** > **Bone Removal Head**
 Bone Removal Body	Masks out bone structures The applied algorithm is optimized for thorax or abdomen regions. Available for VRT and MIP images **Top left corner menu** > **Bone Removal Body**

Bone Removal Leg	Masks out bone structures The applied algorithm is optimized for extremities. Available for VRT and MIP images **Top left corner menu** > **Bone Removal Leg**
Structure Isolation	Allows you to refine bone removal algorithm results You can isolate specific bone structures or non-bone structures. Available for VRT and MIP images **Top left corner menu** > **Structure Isolation**
Edit Bone Removal	Allows you to refine bone removal algorithm results You can mask out bone structures or display non-bone structures again. Available for VRT and MIP images **Top left corner menu** > **Edit Bone Removal**
Bone Opacity	Allows you to define the transparency of bone structures Available for VRT and MIP images **Top left corner menu** > **Bone Opacity**
Table Removal	Allows you to remove irrelevant structures resulting from patient tables or head holders Available for VRT and MIP images **Top left corner menu** > **Table Removal**

Region Growing	Allows you to interactively segment structures such as vessels, bones or organs. Available for VRT, MPR and MPR thick images **Top left corner menu** > **Region Growing** Keyboard shortcut: **X**
Anatomy Visualizer	Allows you to use several masking layers to show or hide the masked structures. Available for VRT, MIP, MPR and MPR thick images **Top left corner menu** > **Anatomy Visualizer**

Tools for restricting the volume

You have tools to mask out structures or parts of a volume.

Top left corner menu	
Punch	Punches out an area of the volume (by keeping the inner or the outer part for reconstruction) With **Reset Visibility Mask**, the visibility mask can be reset. It can be accessed by clicking the settings icon. **Top left corner menu** > **Punch** Keyboard shortcut: **P**
Clip Plane	Hides the volume behind a cutting plane With **Show Clip Plane Graphics**, the clip plane graphics can be turned on or off and with **Change Orientation**, the clip plane orientation can be changed. They can be accessed by clicking the settings icon. **Top left corner menu** > **Clip Plane**

Clip Plane Slab	Only shows the volume inside a slab With **Slab Thickness**, the initial slab thickness can be changed and with **Change Orientation**, the clip plane orientation can be changed. They can be accessed by clicking the settings icon. **Top left corner menu** > **Clip Plane Slab**
Clip Box	Restricts the volume to a cuboid With **Clip Box Graphics**, the clip box graphics can be turned on or off. It can be accessed by clicking the settings icon. **Top left corner menu** > **Clip Box**

Tools to create new series

To create different types of ranges, the system offers you the following tools:

Top left corner menu	
Ranges	Allows you to select a range tool **Top left corner menu** > **Ranges**
Parallel Ranges	Allows you to define parallel slices of the volume for reconstruction **Top left corner menu** > **Ranges** > **Parallel Ranges**
Radial Sliced Ranges	Allows you to define slices of the volume for reconstruction that are arranged around a pivot **Top left corner menu** > **Ranges** > **Radial Sliced Ranges**
Radial Ranges	Allows you to define radial views of the volume for reconstruction **Top left corner menu** > **Ranges** > **Radial Ranges**

Curved Ranges	Allows you to define parallel curved slices of the volume for reconstruction **Top left corner menu** > **Ranges** > **Curved Ranges**
Spine Ranges	Allows you to define slices perpendicular to spinal cord **Top left corner menu** > **Ranges** > **Spine Ranges**
Vascular Ranges	Allows you to create series of images around or along the CPR **Top left corner menu** > **Ranges** > **Vascular Ranges**
Edit Result	Allows you to edit ranges series that were acquired with PACS Ready **Top left corner menu** > **Edit Result**
Save as Series	Allows you to save the series which is displayed in the selected segment as a new series **Top left corner menu** > **Save as Series**

6.2.7 Tools to perform advanced evaluation

For advanced evaluations in MM Reading, the system offers you the following tools in the **Tool Gallery**:

Time Curve evaluation	
Time Curve	Allows you to perform a dynamic evaluation by measuring signal changes over time

Time ROI	Measures the signal intensity for a selected region of interest
Vascular evaluation	
Define Vessel	Activates the vessel definition mode to segment vessels in CT and MR series. The tool allows you to display vessels in curved views and evaluate the vessels accordingly (for example, measuring vessel diameter or vessel length)
Show Vessel	Displays or hides the defined vessels in all image segments
Heart Isolation	Removes surrounding tissue to isolate the heart in a CT cardiac series
Coronary Tree	Allows you to isolate the coronary arteries and the ascending aorta in a CT cardiac series
Oncological evaluation	
Assisted Perpendicular	Allows you to draw perpendicular distance lines to perform diameter measurements and thus, to evaluate lesions

MR calculation tools	
Subtraction	Allows you to subtract different time phases in MR images, MR series, or 4D MR data sets (➔ *Analysis Tools: Performing subtraction*)
Addition	Allows you to add MR images in multiple ways: • Add a constant value to a set of images or to a set of series • Add all images in a series • Add images across different series (➔ *Analysis Tools: Using addition*)
Multiplication	Allows you to multiply a set of MR images or a set of MR series by a constant value. (➔ *Analysis Tools: Using multiplication*)
Division	Allows you to divide MR images in multiple ways: • Divide a set of images or a set of series by a constant value • Divide series by another series • Divide a series by one image (➔ *Analysis Tools: Using division*)
Arithmetic Mean	Allows you to calculate the mean value of the grayscale, to average large intensity differences in the MR images (➔ *Analysis Tools: Calculating the arithmetic mean*)

Motion Correction	Allows you to correct motion artifacts in MR images, for example, for dynamic MR examinations (→ *Correction tools: Performing motion correction*)
ADC & b-value	Allows you to calculate ADC maps from diffusion-weighted images and measurements with various diffusion weightings (b-values). (→ *Analysis Tools: Using ADC & b-value*)
2D Distortion Correction	Allows you to correct geometric distortions which might occur at the edges of MR images with large FoVs (→ *Distortion Correction*)
Image Filter/ Smooth	Allows you to post-process blurred MR images or MR images with strong noise (→ *Image Filter/Smooth*)
MR Composer tools	
New Compose	Allows you to compose MR images, for example, to compose all suitable series of a Whole Spine MR study in one series (for sagittal or coronal images) (→ *Creating composing*)
New Combine	Allows you to combine images, for example, to combine all images in one series (for axial images) (→ *Creating a new combination*)

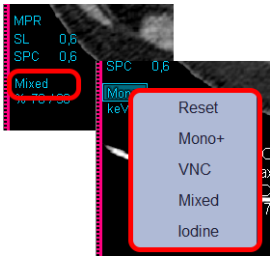
Lung evaluation tools	
CT Lung Change	Allows you to display anatomical changes to lung tissue in CT images across different time points
Lung CAD	Marks suspect lung tissue Allows you to review lung nodule markers proposed by the system. In the **Lung CAD** mini toolbar you can navigate through these markers, create findings out of the markers or reject markers

Spectral Imaging

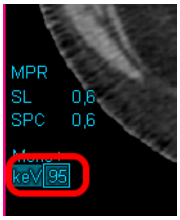
The **MM Reading** workflow supports spectral imaging data (dual energy study) so that you can interactively change different energy levels (Mono+ keV levels) and evaluate certain dual energy representations (for example, VNC, Iodine).

The spectral imaging feature requires Spectral Post-Processing (SPP) image data that can be easily created in *syngo*.via.

These adaptions can be made with interactive image text. The settings are available once a dual energy study (SPP) has been loaded to **MM Reading**.



You switch between the different representations by clicking the corresponding image text, or by right-clicking the image text and choose another representation from the context menu.



The energy levels can be changed by moving the mouse up or down, or by right-clicking and adapting the given value in the corresponding dialog box.

6.2.8 Printing of images

With the print functionality, you can expose images on film at a DICOM printer, or print them on paper on a standard Windows printer.

Printouts can be used for reporting, archiving, documentation, and treatment decisions.

After you have sent images for printing, the film sheets display a print preview of all series or images.

You can send images for printing, for example, from the image area and the **Series** panel.

Depending on the workflow, the printing functionality is either available as a separate workflow step or as a floating window.

- The **Print** workflow step will be available as soon as you have sent images for printing.
- The **Printing** window will open when you click the **Printing** button in the **Viewing** panel of MM Reading.

On the film sheet, you can perform the following actions:

- Modify images

  Change layouts, rearrange the image order, or change the image display
- Modify image text

  Define the amount of image text, that will be shown on the film sheet
- Add annotations and arrows

- Set **Printing Properties**

  Configure automatic functions, select a printer or layout, and set options for the film sheet and the printout

- Print images

  Select a printer, a layout, and the paper size and then start a print job

Print tools in the Viewer

You have several tools to initiate printing.

Favorite Tools panel / floating segments (opened from the Findings panel or the Series panel)	
Print Image	Allows you to send the currently selected image to the film sheet. Keyboard shortcut: **E** or **numpad Enter**
Print Stack	Allows you to send all images of a stack to the film sheet. Keyboard shortcut: **A**
Print Layout	Allows you to send the visible layout of one or more monitors to the film sheet. In multi-monitor setup, press and hold the **Ctrl** key and click a segment on each monitor that should be printed.

6.2.9 Findings panel

The **Findings Assistant** supports the creation and management of structured findings.

You will find this component in the **Findings** panel.

The **Findings** panel can be undocked (opened) and moved to another position on the screen.

Findings

The **Findings** panel provides two ways of creating structured findings:

- You create a non-measurement finding, select a specific finding type, and then start drawing the measurements according to the given placeholder values.
- You start by creating measurements, which are marked as findings, and then create a finding to group them and assign the finding type.

After you have created structured findings, you proceed by specifying the findings attributes, rating and commenting findings, and assigning specific finding types, for example, **Stenosis** finding.

Finding types

Finding types allow for data to be collected, displayed, and annotated in a structured format. Some of the finding types are based on clinical guidelines.

Based on your inputs, you can qualify the finding according to the given standards.

Body regions

Body regions allow you to provide general information about the examined organ or body region and to perform an overall quantification of the findings within the body region. You can assign findings to a default body region. The body region structure that you define in the **Findings** panel is also reflected in the report.

Follow-up findings

For follow-up examinations, the findings of the previous examination are loaded and you can add measurements of the current examination to the same finding. This is helpful for trending findings. All findings are automatically sent to the report editor.

Export of findings

The created findings can be exported directly from the **Findings** panel to PDF or CSV format, to a printer, to clipboard, or as e-mail. The export to e-mail option will only be available if an e-mail program is available on the client.

6.2.10 Findings Navigator

The **Findings Navigator** collects measurements, annotations, and graphical objects marked as findings.

You can review and modify those findings at any time during evaluation. Later, the collected findings can be included into the report.

Depending on the configuration, findings can be created manually or automatically. It also depends on the configuration, if snapshots are available for each finding.

Consider that not all workflows support the **Findings Navigator**.

In multiple workflows, the **Findings Navigator** has been replaced by the **Findings** panel.

6.2.11 Handing over a workflow

If you want to hand over the workflow to another user, simply proceed as follows:

✓ A workflow has been started and images are loaded for reading.

1 Click the **Save and Pause** icon to close reading and save unsaved results.

If you want to discard all results from this session, click the **X** icon in the Patient tab to open the **Close Patient** dialog box and click **Discard Changes**.

2 Inform the receiving user that the workflow is now ready to be opened in full access mode.

6.2.12 Saving a workflow

When you save a workflow, the workflow is closed and results are saved in the Short Term Storage (STS).

When re-opening the workflow, the workflow is opened at the position it was saved.

Result series are indicated by an icon next the series name in the **Series** panel. This icon will remain in this section upon re-opening the workflow.

Saving a workflow does not mean to archive a workflow!

If you do not archive the workflow within eight days, the saved workflow will be cancelled and all results of the workflow are ready to be deleted!

✓ A workflow has been started and images are loaded for reading.

◆ Click the **Save and Pause** icon.

The workflow is closed.

In the **Workflow Status** column of the **Patient Browser**, the state of the workflow is set to **Saved**.

6.2.13 Discarding a workflow

When you discard a workflow, the workflow is closed, and all unsaved results are lost.

✓ A workflow has been started and images are loaded for reading.

1 In the Patient tab, click the **X** icon.

The **Close Patient** dialog box is opened.

2 Click **Discard Changes** to close reading and discard all results from this session.

The workflow is closed and unsaved results are lost.

In the **Workflow Status** column of the **Patient Browser**, the state of the workflow changes to **Ready**.

6.2.14 Saving and archiving a workflow

When you complete a workflow, the workflow is closed and all unsaved results are saved to the Short Term Storage (STS). The results are sent to the configured archive.

The result series are saved as DICOM series. If you open the workflow again, the result series will be displayed in the **Current**, **Later**, or **Prior** section of the **Series** panel.

✓ A workflow has been started and images are loaded for reading.

◆ Click the **Save and Send** icon.

The workflow is closed.

In the **Workflow Status** column of the **Patient Browser**, the state of the workflow is set to **Completed**.

In case of an error, after 15 minutes, all storable results are saved and the workflow is set to **Completed**.

The error is reported in the log file.

6.3 About *syngo*.via Reporting

To make reporting easier for the radiologist, *syngo*.via workflow templates are assigned to report templates (1).

While synchronizing selected *syngo*.via data with the *syngo*.via **Report Editor**, all relevant data are automatically imported to the appropriate report section (2). The radiologist then only customizes the report content where needed.

If a workflow is completed, the report is automatically sent to the *syngo*.via Short Term Storage (STS). With *syngo*.via Reporting the radiologist can save or print a report, or send it to a connected information system, such as a HIS or RIS (3).

Findings constitute the core of a report. The radiologist creates findings in *syngo*.via and transfers them to the **Report Editor**. The presentation of findings in a report depends on the sections used in the assigned report template. The sections specify the findings renderings. If a finding is modified in *syngo*.via, the corresponding value and snapshot in the report are updated immediately. The radiologist can also manipulate findings or change the editing state in the **Report Editor** to continue editing a report later on. The range of functions depends on the license:

- With the basic feature set, the radiologist can continue working on a report, which was completed. When you restart working on the same workflow, you will be asked if you want to continue working on the previously created report or if you want to create a new report.
- With the appropriate license, the radiologist can select, if a report shall remain editable when completing a workflow. You can use the report of a completed workflow and continue reporting with data of a different workflow.

Please note that various reporting features are only available with the appropriate license. For licensing information please refer to the *syngo*.via Data Sheet.

6.3.1 Report Editor

The **Report Editor** is designed to create, customize, distribute, and print reports. These reports or parts of the reports can be transferred via copy & paste to clinical reports in a RIS or to other external reporting solutions. When opened, the **Report Editor** synchronizes automatically with running examinations or reading tasks.

The **Report Editor** is displayed as a floating window so as not to interfere with your current work in *syngo*.via. You can move this window to any position or maximize it. The last position of the window is re-established when re-starting the **Report Editor**.

When you switch between cases during reporting, the **Report Editor** is kept open until you explicitly close it.

This chapter describes the following aspects:

- Screen layout of the **Report Editor**
- Starting the **Report Editor**

  (→ Page 131 *Starting the Report Editor*)
- Main features of the **Report Editor**

In the **Report Editor**, the template language is defined by the *syngo*.via user interface language. With the appropriate license you can change the report language manually, if the content is available in the desired language.

Date and time format within the **Report Editor** are defined by the specified date and time format of your client operating system.

Starting the Report Editor

If configured in the **Reporting Configuration** settings, the **Report Editor** starts automatically.

✓ A procedure is loaded in a workflow.

◆ In the **Patient** tab, click the **Report Editor** icon.

The **Report Editor** opens. The selected finding data are automatically imported.

You can also start the **Report Editor** with the **Alt** + **R** keys.

Please note that a report is only generated, once the **Report Editor** has been opened.

6.4 About the export of data (images)

Exporting data in DICOM format

Using the **Export Data** dialog box, you can export selected data in DICOM format.

You can export to the following locations:

- CD/DVD

  In case there is no CD/DVD drive available at your system, an image file (*.iso) will be created, and can be saved to a local drive.

- File system
- DICOM nodes

  For example, to the teamplay cloud, or for archiving in the PACS

  (→ *Configuration of remote DICOM nodes*)

First, you select the data to be exported in the **Patient Browser** or in the Viewer:

(→ Page 133 *Exporting from the Patient Browser*)

(→ Page 134 *Exporting from the Viewer*)

In the **Export Data** dialog box, you can refine the data selection and select the export target (➔ Page 136 *Screen layout of the Export Data dialog box*):

For configuration, see:

(➔ Page 137 *Settings for data export*)

(➔ *Defining media writing profiles*)

Differences in exporting conditions:

- From the **Viewer**: Data of the **currently active** evaluation can be transferred as one data set.
- From the **Patient Browser**: **Any** data (for example, DICOM Structured Reports, Encapsulated PDFs, segmentations) can be transferred as one data set.

When you close the **Export Data** dialog box without starting the export, the previous data selection is retained. This allows you to collect data from the **Patient Browser**, from segments and from the **Series** panel for a single export operation.

Exporting data in non-DICOM format

You can export images and movies in non-DICOM image formats (such as .JPG, .AVI, and .WMV) to display them also with common viewers and media players.

These images and movies are only intended for presentation purposes.

6.4.1 Exporting from the Patient Browser

From the **Patient Browser** **any** data can be transferred as one data set such as:

- the complete data set of a patient

  To export the complete data set of a patient, you have to select all studies of the patient.

- selected studies and series
- single images or image stacks

  These images are transferred as secondary captures including all evidence objects, such as measurements.

1 In the **Results**, **Series** or **Instances** list of the **Patient Browser**, select the series, studies or images you want to export.

  In the Patient View, you can only select studies. You cannot select patient entries.

2 On the toolbar of the **Patient Browser**, click the **Export** icon.

  The **Export Data** dialog box opens and the data you have selected is displayed.

3 Adjust the relevant settings in the **Export Data** dialog box.

  (→ Page 136 *Screen layout of the Export Data dialog box*)

6.4.2 Exporting from the Viewer

From the Viewer, the following data of the **currently active** evaluation can be transferred as one data set:

- Selected studies and series

  (→ Page 135 *Exporting studies and series from the Series panel*)

- Single images or image stacks

  These images are transferred as secondary captures including all evidence objects, such as measurements.

  (→ Page 135 *Exporting images from the image area of the Viewer*)

  (→ Page 135 *Exporting images from a floating segment*)

  (→ Page 136 *Exporting images from the Findings Navigator*)

Exporting studies and series from the Series panel

The following steps allow you to export selected studies and series from the **Series** panel.

✓ The **Series** panel is open and the **List** view is switched on.

1 In the **List** view, select a series or a study.

2 Right-click and choose **Export Series** from the context menu.

The **Export Data** dialog box opens and the selected data is displayed.

3 Adjust the relevant settings in the **Export Data** dialog box.

(➔ Page 136 *Screen layout of the Export Data dialog box*)

Exporting images from the image area of the Viewer

The following steps allow you to transfer single images or image stacks which are displayed in the image area of the **Viewer**:

1 In the **Viewer**, select a segment (multiple selection with the **Ctrl** key).

2 In the **Favorite Tools** panel, click the **Export Image** icon to export a single image.

– or –

Click the **Export Image** palette icon and select the **Export Stack** icon to export entire series.

With the **Export Stack** function, you can only export stacks of 2D original images. For all other display types, the icon is dimmed.

The **Export Data** dialog box opens and the selected data is displayed.

3 Adjust the relevant settings in the **Export Data** dialog box.

(➔ Page 136 *Screen layout of the Export Data dialog box*)

Exporting images from a floating segment

The following steps allow you to specify which images of a finding or series are exported:

1 In the **Findings** panel, double-click a snapshot icon.

– or –

In the **List** view of the **Series** panel, double-click a series.

The images are displayed in a floating segment.

2 In the floating segment, move the mouse over the watermark to display the mini toolbar.

3 On the mini toolbar, click the **Export Image** icon to export a single image.

– or –

Click the **Export Stack** icon to export entire series.

The **Export Data** dialog box opens and the selected data is displayed.

4 Adjust the relevant settings in the **Export Data** dialog box.

(→ Page 136 *Screen layout of the Export Data dialog box*)

Exporting images from the Findings Navigator

The following steps allow you to export the images of a finding.

1 In the **Findings Navigator**, select the desired findings (multiple selection with the **Ctrl** key).

2 Right-click and choose **Export Image(s)** from the context menu.

The **Export Data** dialog box opens and the selected data is displayed.

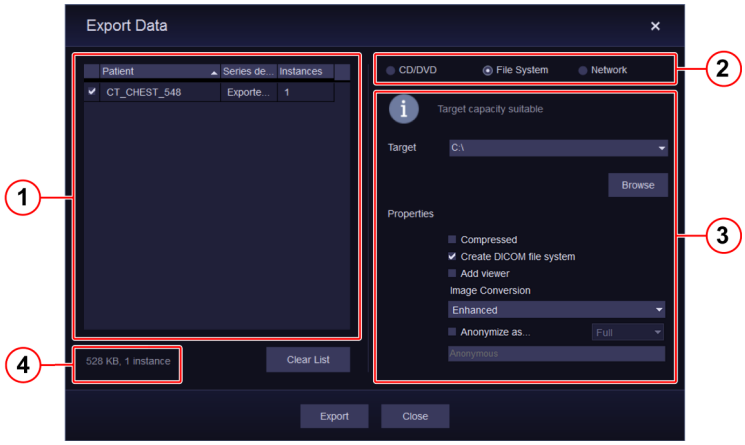
3 Adjust the relevant settings in the **Export Data** dialog box.

(→ Page 136 *Screen layout of the Export Data dialog box*)

6.4.3 Screen layout of the Export Data dialog box

In the **Export Data** dialog box, you can adjust settings for the export of data to the local hard disk, to removable media, or to network nodes and start the export job.

The **Export Data** dialog box comprises the following sections:



(1) Data selection area

Contains a list of data available for export and check boxes for marking data to be exported

(→ Page 134 *Exporting from the Viewer*)

(→ Page 133 *Exporting from the Patient Browser*)

The **Clear List** button removes all entries in the data selection area

(2) Export target area

Enables you to select whether to export data to a CD or DVD, to the file system, or to a network node

(3) Export settings area

Changes with the selected export target option. You can select the target.

(→ Page 137 *Settings for data export*)

(4) Amount of selected data to be exported

6.4.4 Settings for data export

In the operations area of the **Export Data** dialog box, you can adjust all relevant settings for the selected export target.

(→ Page 136 *Screen layout of the Export Data dialog box*)

In the following, you find an overview of all parameters to be adjusted when exporting selected data to **CD/DVD**, your **File System**, or other DICOM nodes in the **Network**.

- For exports to **CD/DVD**, parameters are set by the media profile. The parameters are displayed in the **Export Data** dialog box but cannot be modified. To change these settings, you have to adapt the media profile.
- For exports to the **File System**, parameters can be set and selected in the **Export Data** dialog box.

syngo.via can display incoming images that have DICOM grayscale Presentation States associated. The image display obeys the presentation parameters (like windowing, zoom/pan, annotations, measurements) described by these Presentation States.

For the export of such DICOM data to media or local file system, it is possible to choose between two output format types:

- **Interoperability**: the graphics on the source images are transformed into an interchangeable output format. Note that this output format might not have the best quality.
- **Enhanced**: the graphics format on the output images is the same as on the input, which ensures an optimal image quality.

If you selected **CD/DVD** as export target, the following parameters are available:

Target	Determines the drive of the media device

Action	When exporting data to CD/DVD, an ***.iso** image file containing a copy of the original CD/DVD is generated and stored on your local hard disk. In the **Export Data** dialog box, you can select your preferred option for proceeding: • **Burn** Automatically burns the data on CD/DVD on the basis of the ***.iso** image file. This option is selected by default. • **Open with Explorer** Opens the ***.iso** image file in the Windows Explorer, for example, to first view the file before burning the data on CD/DVD. Note: The last two ***.iso** image files are stored on your computer. **Image Conversion** • **Interoperability**: assures that the data is "understood" by any DICOM node, but enhanced features cannot be guaranteed. • **Enhanced**: preserves the enhanced features of the source images.
Media label	User-defined supplement of the CD/DVD label Note: Only letters, numbers and underscore are allowed, maximum 6 characters. The user-defined media label is extended with the current date and time (YYMMDDhhmm).

Profile	Determines the profile of the target medium, for example, Demo Medium, Patient Medium, or Standard (and private profiles) An export profile defines settings such as image compression, DICOM file system, addition of a viewer, and image conversion. You can add, modify, and delete these profiles. (→ *See: Defining media writing profiles*).

If you selected **File System** as export target, the following parameters are available:

Target capacity	Shows if the capacity of the target medium is suitable for the current export job, or not
Target	Path in the file system (including shared network drives and USB devices). For easy navigation use the icons.

Properties	Export options: • **Compressed** Data is compressed before export • **Create DICOM file system** A file directory according to DICOM standards is created • **Add Viewer** A DICOM viewer software is added for viewing of the exported data • **Image Conversion** The options are: – **Interoperability**: assures that the data is "understood" by any DICOM node, but enhanced features cannot be guaranteed. If **Add Viewer** is selected, it cannot be modified. – **Enhanced**: preserves the enhanced features of the source images. • **Anonymize as** Data are anonymized – **Full**: DICOM-conform anonymization – **Reduced**: DICOM-conform anonymization. This option keeps UIDs, patient characteristics and date. – **Service**: Anonymization for service purposes. Enter the name you want to have displayed as patient name.

If you selected **Network** as export target, the following parameter is available:

Target	Allows you to establish a connection to defined DICOM network nodes.

6.5 About the archiving of data (images)

If your system is connected to a PACS or a corresponding DICOM node, relevant data can be sent to an archive automatically or manually.

CAUTION

Long-term archiving is not provided with this system.

Loss of data (including reports) can occur if data have not been transferred to a long-term archive.

- Verify that local data has been transferred successfully to a long-term archive, before deleting it.
- Check the status bar for archiving failures (indicated by an error icon).
- Check the **Job View** to locate the error and take appropriate measures (for example, re-start failed archive jobs).

At least one DICOM node must be configured as a PACS.

For the archiving of data, you have the following general options:

- Globally all relevant data are archived automatically according to specific archiving rules
- Individually mark data in the **Patient Browser** for automatic archiving (→ Page 143 *Selecting data for archiving*)
- Individually select data in the **Patient Browser** to send them to the archive manually

The archive status of objects may change to "Archive failed" after archiving.

If the following factors are combined, the archive status of objects may change to **Archive failed** after archiving:

- low bandwidth
- lossy compression
- disabled storage commitment

To avoid this situation please check the following options:

- **Transfer format optimizations** is set to **Preference to uncompressed format (Optimize CPU Load)**
- **Default Media Compression** priority contains only lossless compression algorithm.
- **Storage Commitment** is enabled.
- **Transfer Syntaxes** are not excluded on SCU and SCP side.

Depending on the configuration, data is archived with the following settings:

- **Interoperability**: the graphics on the source images are transformed into an interchangeable output format. Note that this output format might not have the best quality.
- **Enhanced**: the graphics format on the output images is the same as on the input, which ensures an optimal image quality.

You may also be interested in the following topics:

- View information on the archive status of data
- Check for successful archiving (→ Page 145 *Checking for successful archiving*)
- Prevent data from being archived

6.5.1 Selecting data for archiving

In the **Patient Browser**, you can mark data for archiving.

Data will be archived by the next archiving session as defined in the ***syngo*.via Administration Portal**. If required, you can trigger archiving of selected data manually.

Information on the archive status of data is shown in the **Study Archived**, **Series Archived**, and **Archive Status** columns of the **Patient Browser**.

✓ You have the appropriate user rights.

CAUTION

Data transfers between systems are not guaranteed.

Loss of data if data are deleted locally before they have been successfully transferred to another system.

- Only systems and communication protocols supporting Storage Commitment guarantee successful transfer to another system.
- In other transfers, it is necessary to verify the correct arrival of the data at the remote system itself.
- Do not delete local data until the transfer to the remote system is verified.

1. In the **Results** list of the **Patient Browser**, select the required studies, series, or instances.
2. Right-click and choose **Set Archive state** > **Mark for Archiving** from the context menu.

   Archiving will start as defined in the ***syngo*.via Administration Portal**.

You may also be interested in the following:

- (→ *Configuring archives*)
- (→ Page 142 *About the archiving of data (images)*)

6.5.2 Checking for successful archiving

In the **Patient Browser**, you can search for data that was neither archived nor deleted.

✓ The **Patient Browser** is open.

1 Click **More Filters** to open the **More Filters** area.

2 In the **Study Archived** field, select "Queued for archiving".

In the **Workflow Status** field, select "Completed".

If required, you must add these fields to the **More Filters** dialog box first.

3 In the **Results** list, search for archivable data older than 48 hours.

4 Check why this data was neither archived nor set to "not to be archived".

Data in status "Queued for archiving" are waiting for a storage commit response from the archive. Depending on the archive, this process can take a couple of days.

Also check whether there are workflows in status "Saved" and complete them. Otherwise the results may be automatically deleted as they are in status "Not to be archived".

You may also be interested in the following:

(→ *Configuring archives*)

(→ Page 143 *Selecting data for archiving*)

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